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(54) **IMAGE FORMING APPARATUS**

(56) **References Cited**

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U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **18/494,053**

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Primary Examiner — Jessica L Eley

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(30) **Foreign Application Priority Data**

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(57) **ABSTRACT**

(51) **Int. Cl.**
G03G 21/16 (2006.01)
G03G 15/16 (2006.01)

(52) **U.S. Cl.**
CPC **G03G 21/1633** (2013.01); **G03G 15/165**
(2013.01); **G03G 2215/00628** (2013.01);
G03G 2221/169 (2013.01)

(58) **Field of Classification Search**
CPC G03G 21/1633; G03G 21/1647; G03G
21/1638; G03G 2215/00544
See application file for complete search history.

An image forming apparatus includes an image forming unit, a casing, a door, a connecting member, a guide member, and an operating portion. The operating portion moves, through an operation by a user when the door is at a first opened position, from a regulating position where the connecting member is regulated from moving from a first guide to a second guide, to an allowing position where the connecting member is allowed to move from the first guide to the second guide. When the operating portion is at the regulating position, the door is configured to be retained at the first opened position by engagement of the connecting member and the first guide. When the operating portion is at the allowing position, the door is configured to be retained at the second opened position by movement of the connecting member from the first guide to the second guide.

14 Claims, 32 Drawing Sheets

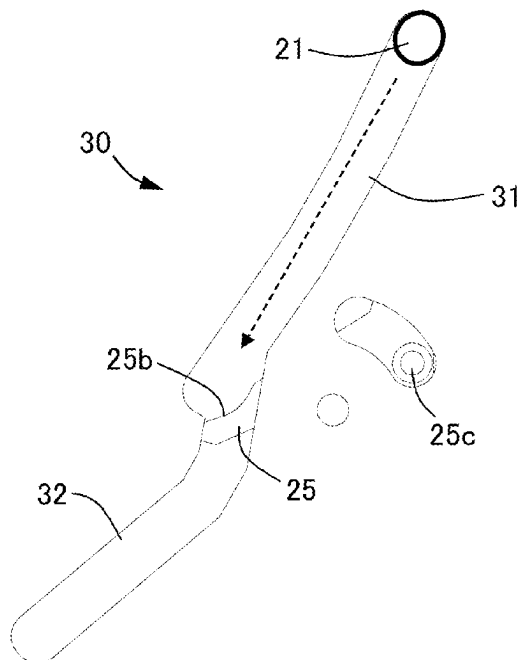


FIG. 1

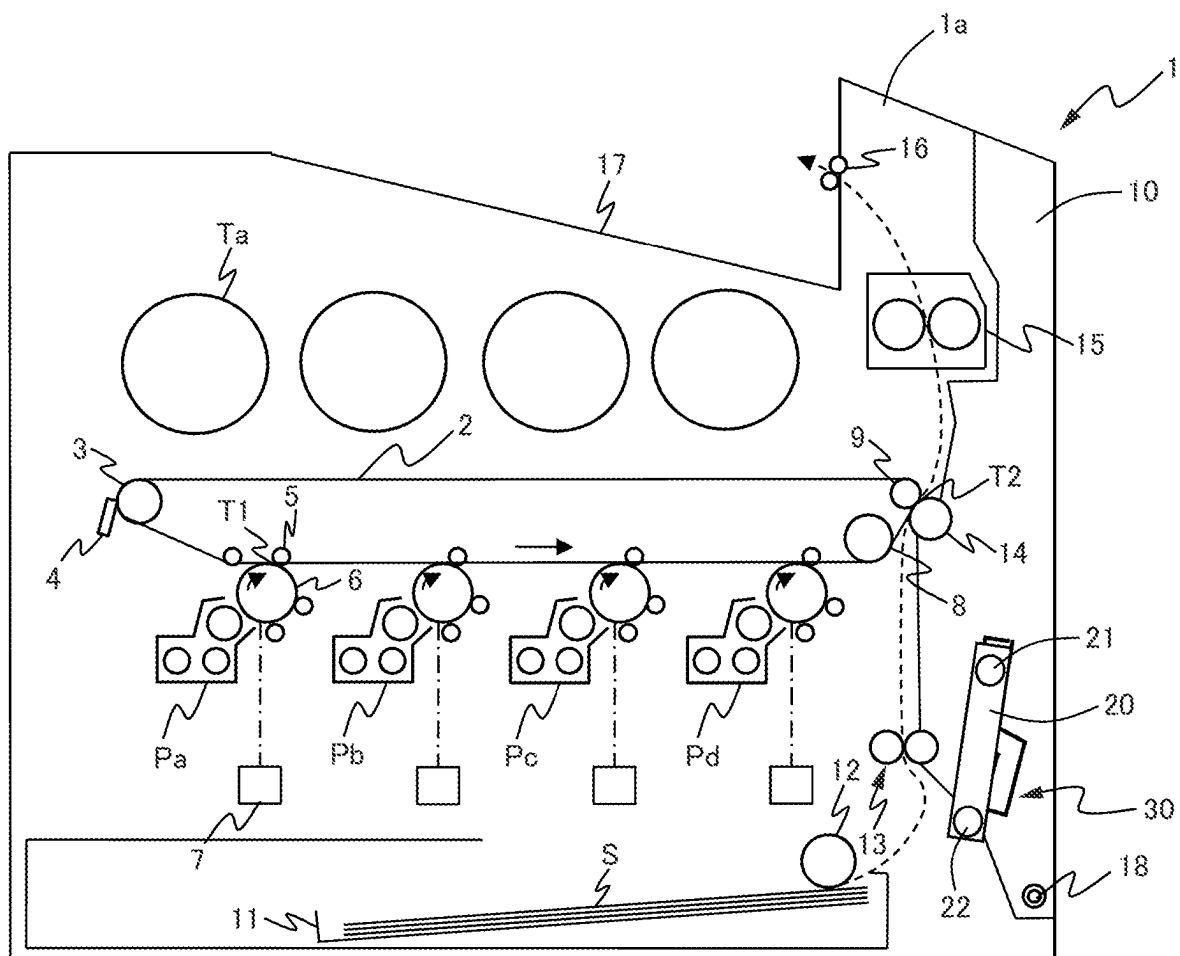


FIG.2A

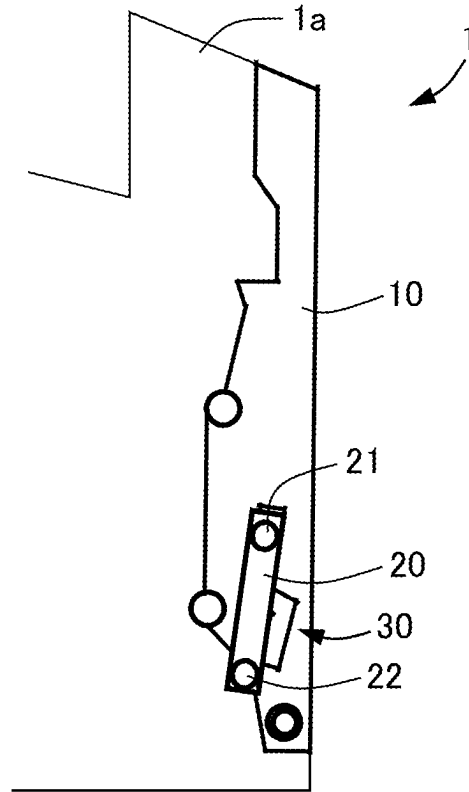


FIG.2B

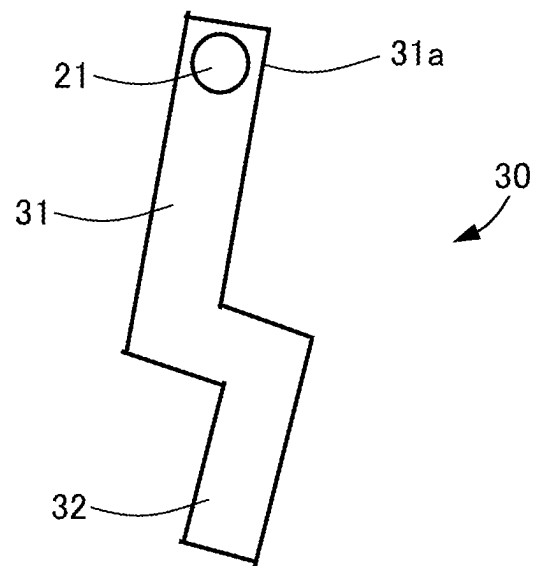


FIG.3A

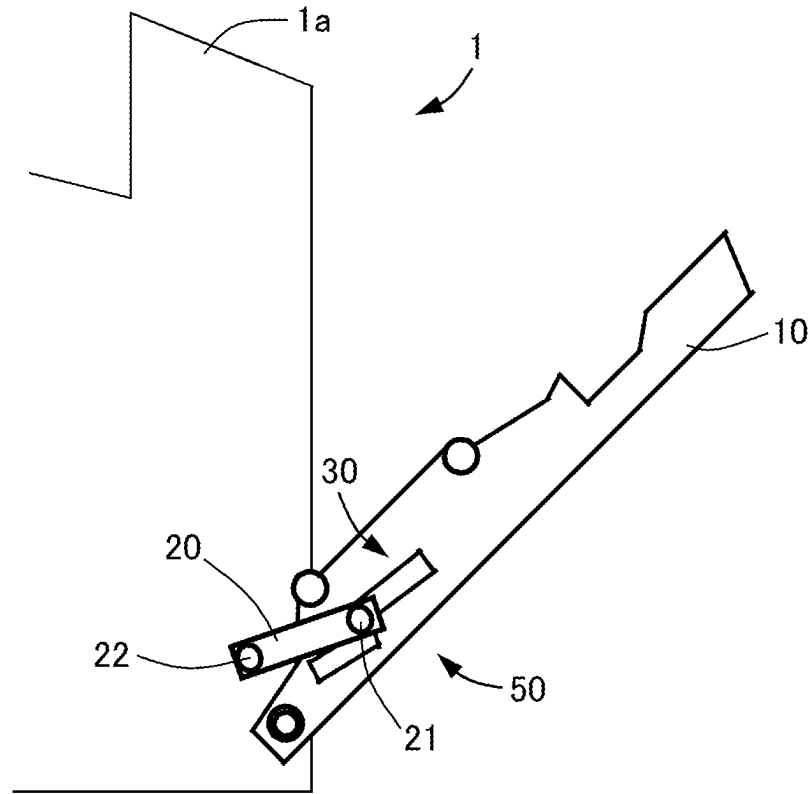


FIG.3B

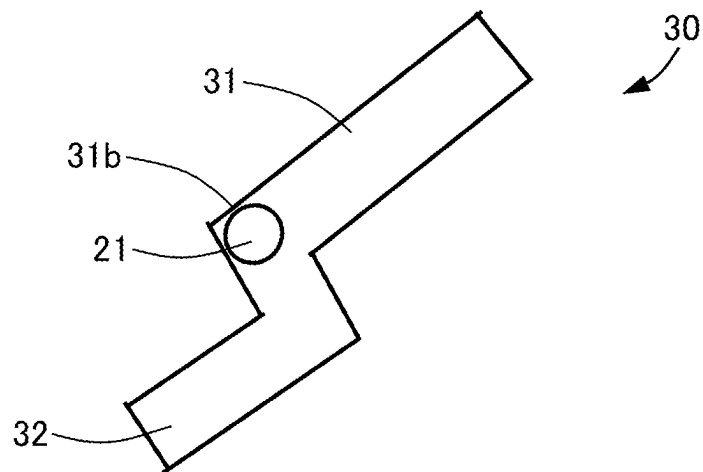


FIG. 4A

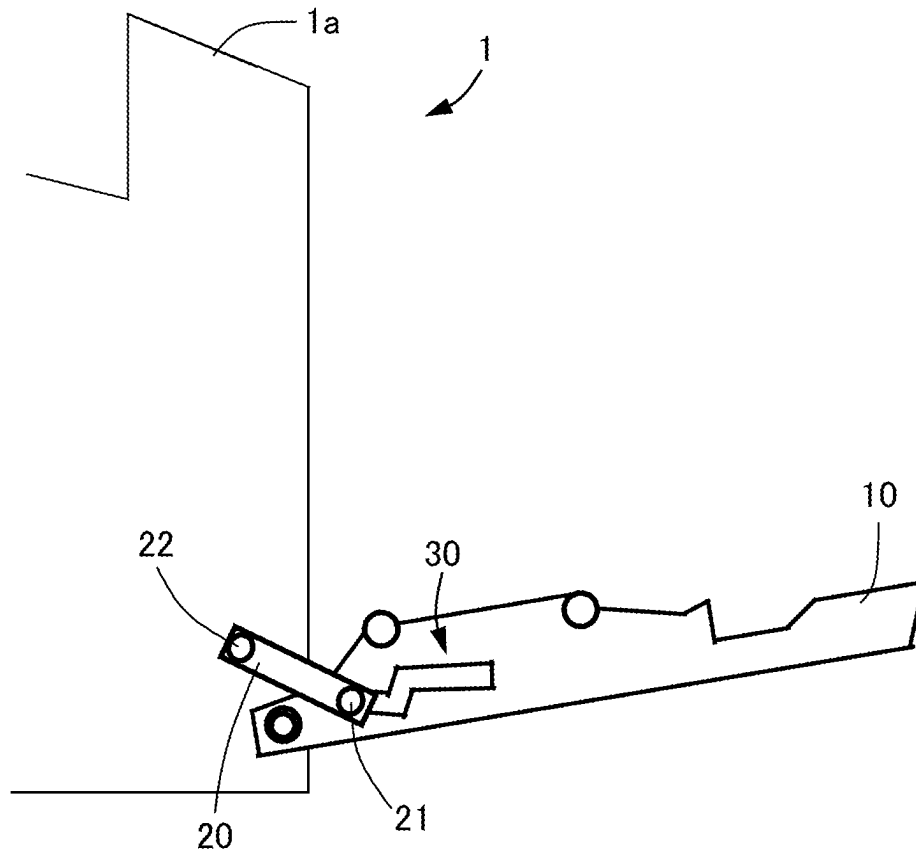


FIG. 4B

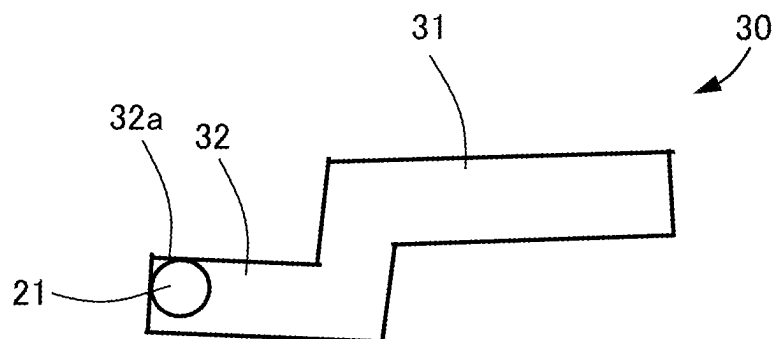


FIG.5A

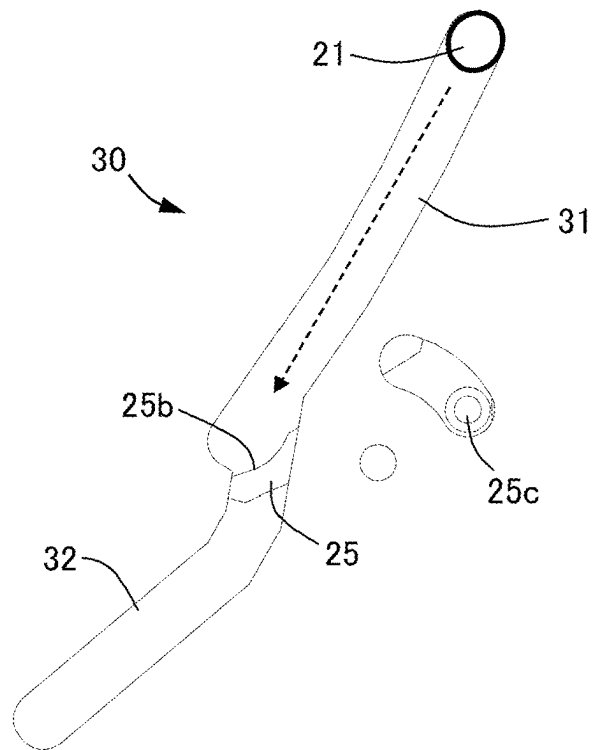


FIG.5B

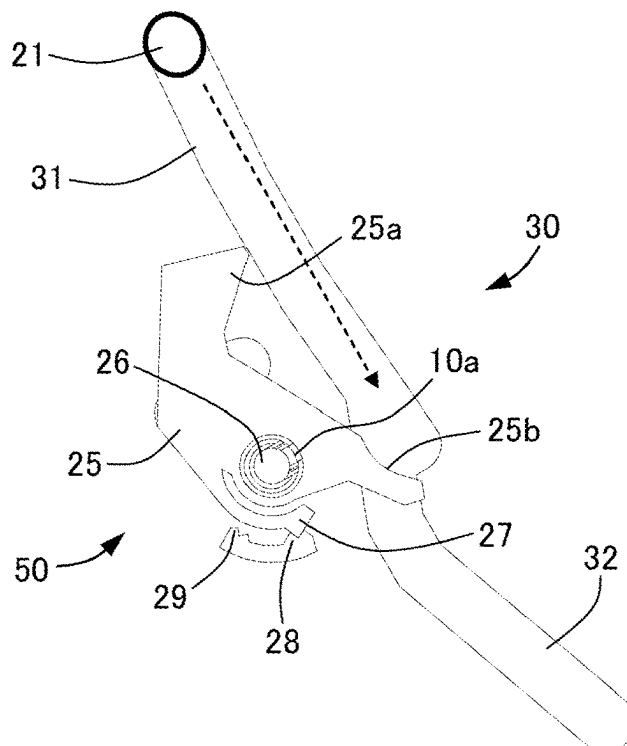


FIG.6A

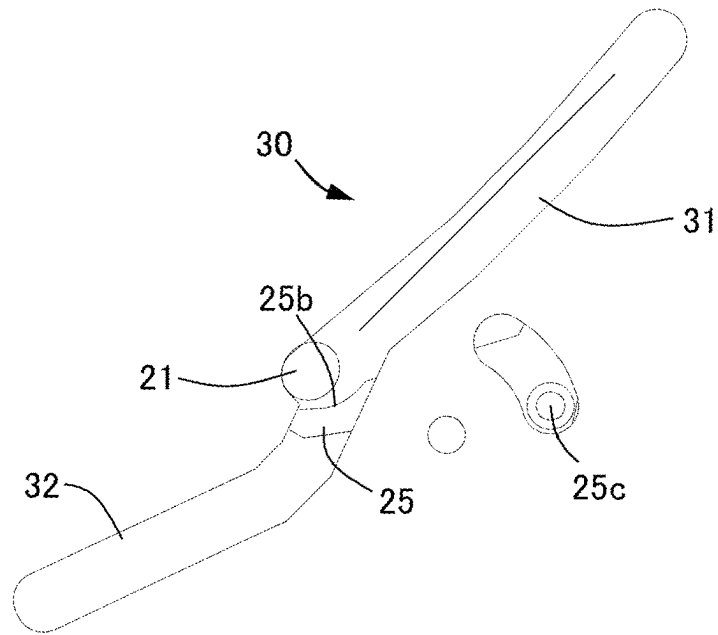


FIG.6B

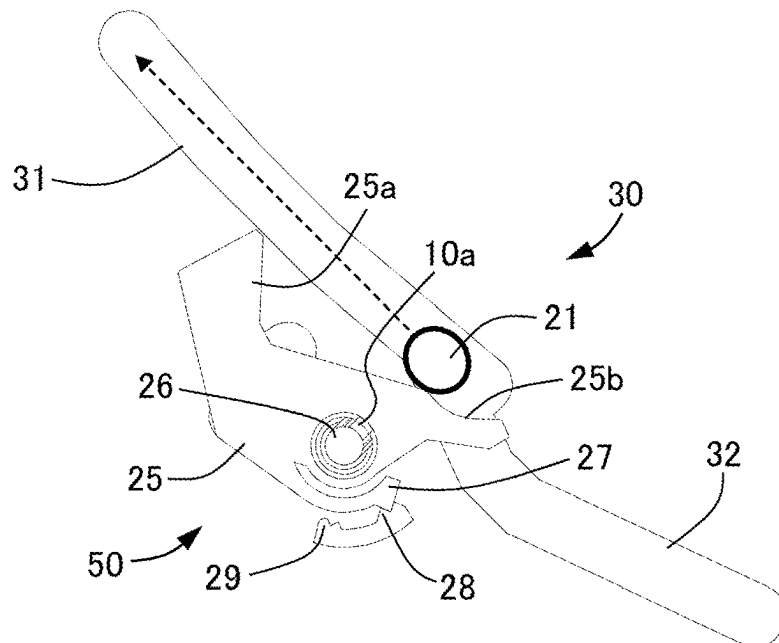


FIG.7A

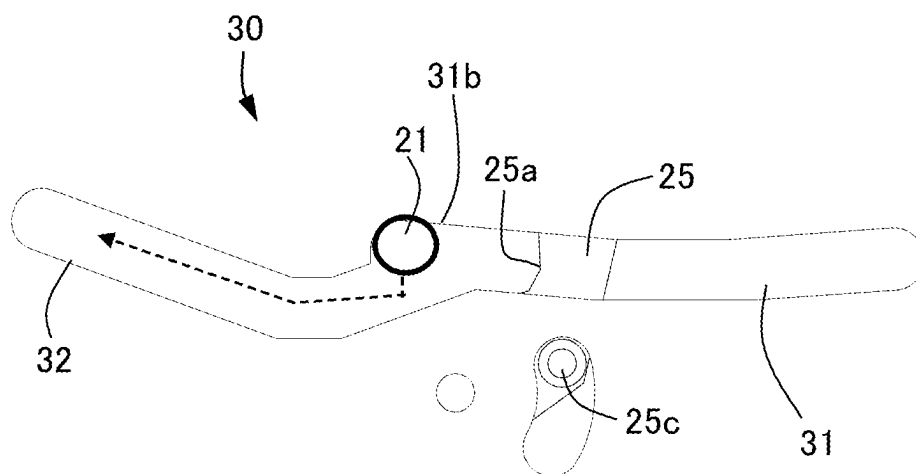


FIG.7B

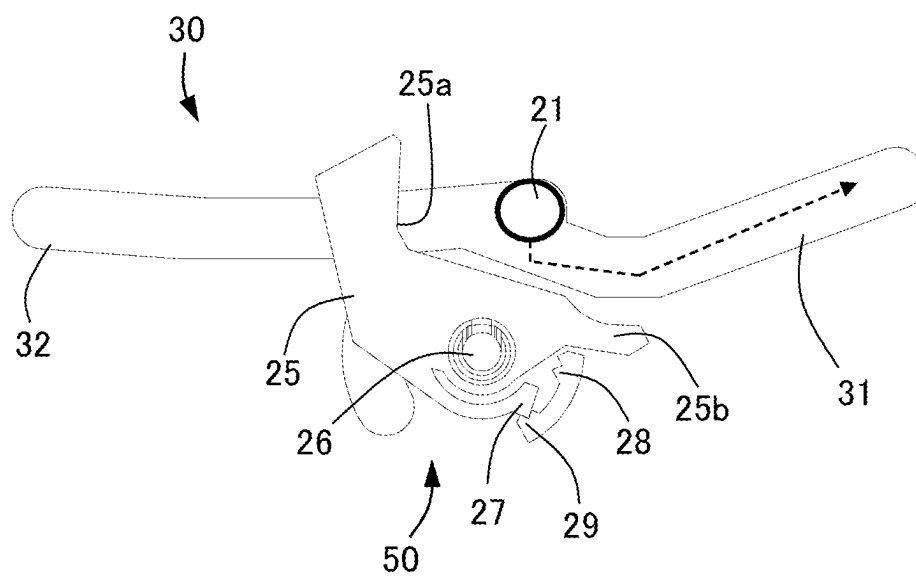


FIG.8A

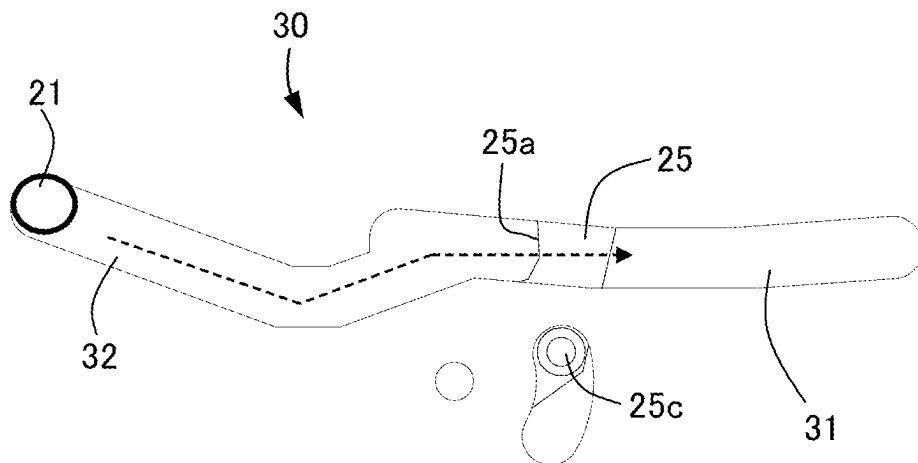


FIG.8B

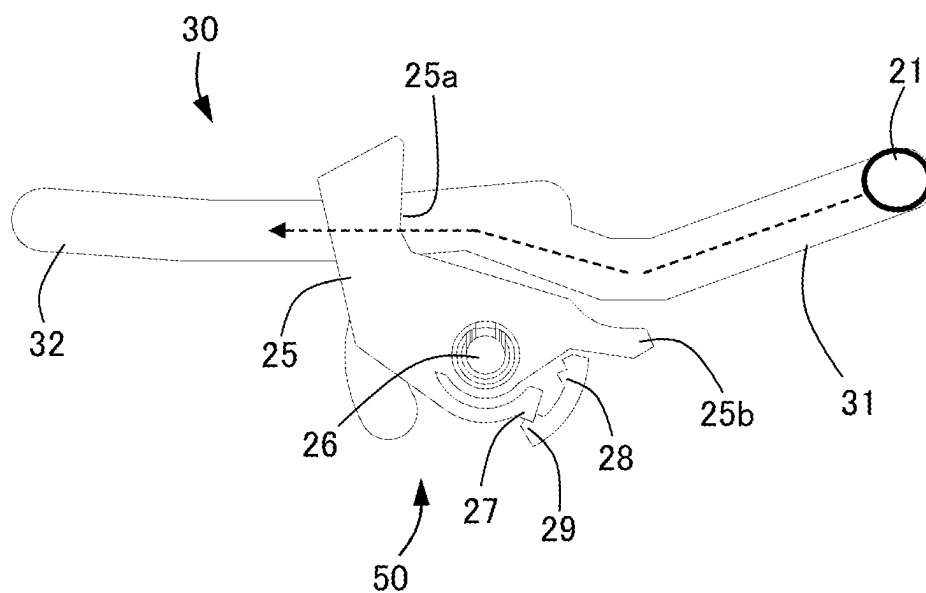


FIG.9A

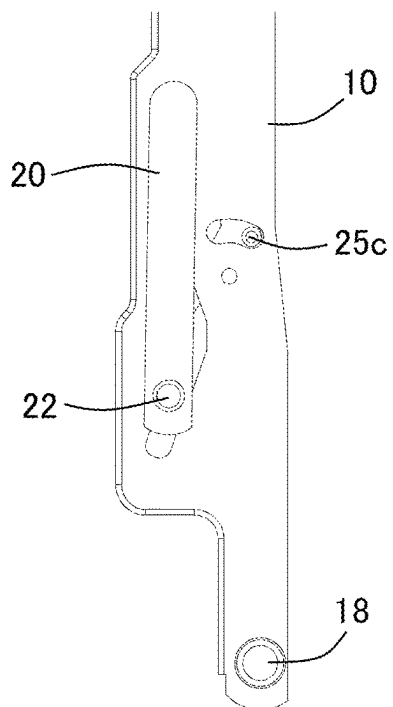


FIG.9B

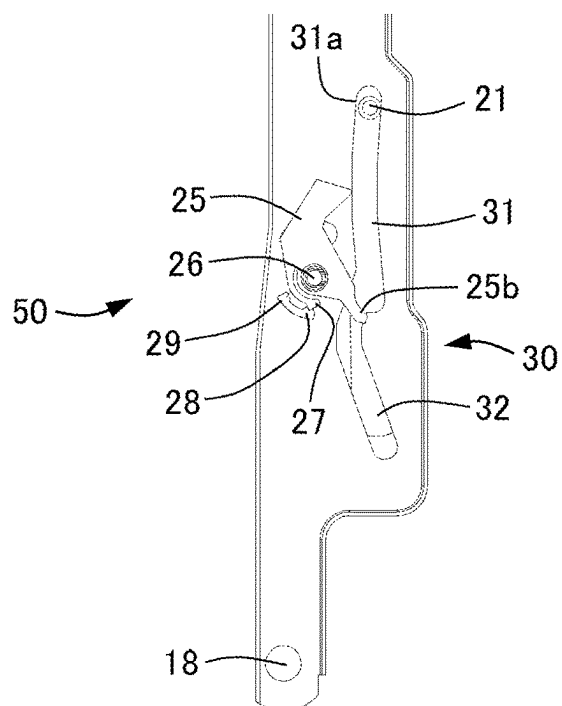


FIG.10A

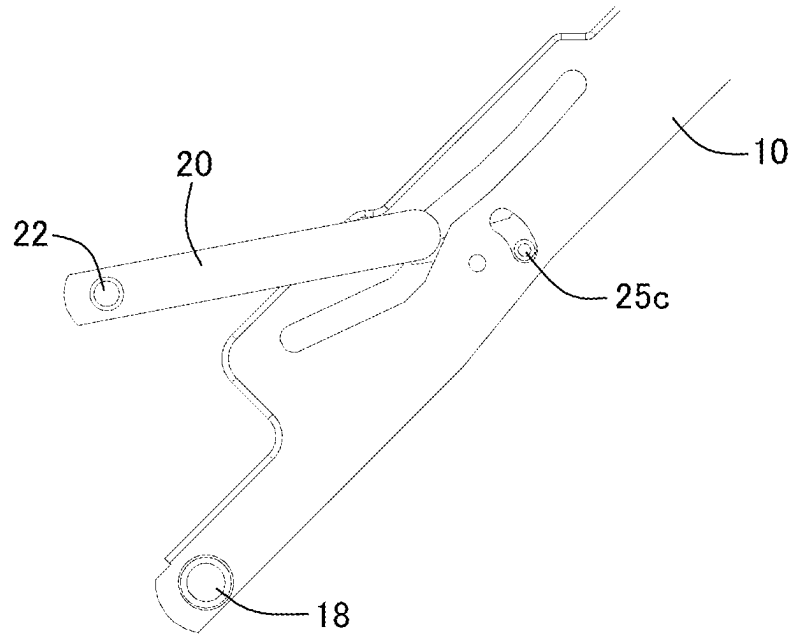


FIG.10B

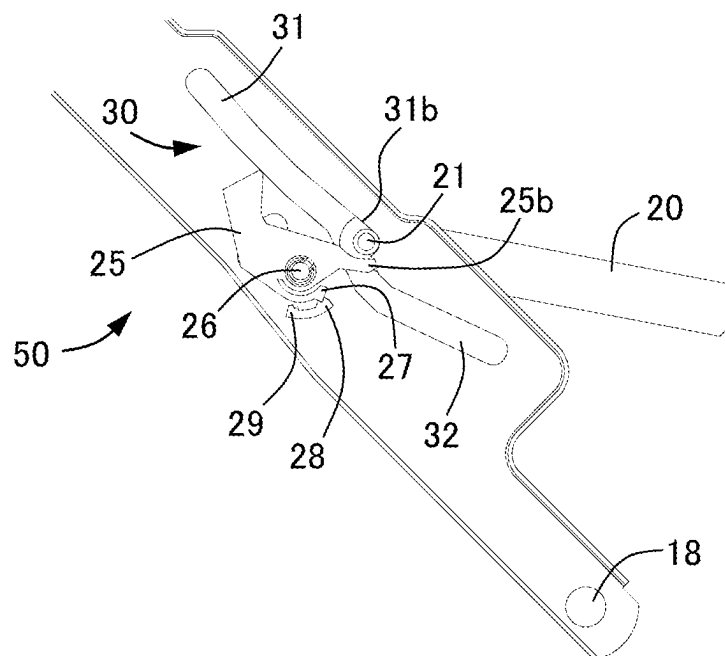


FIG.11A

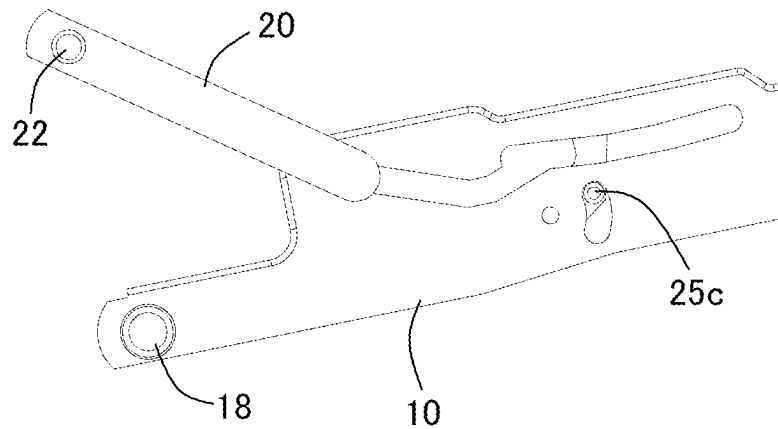
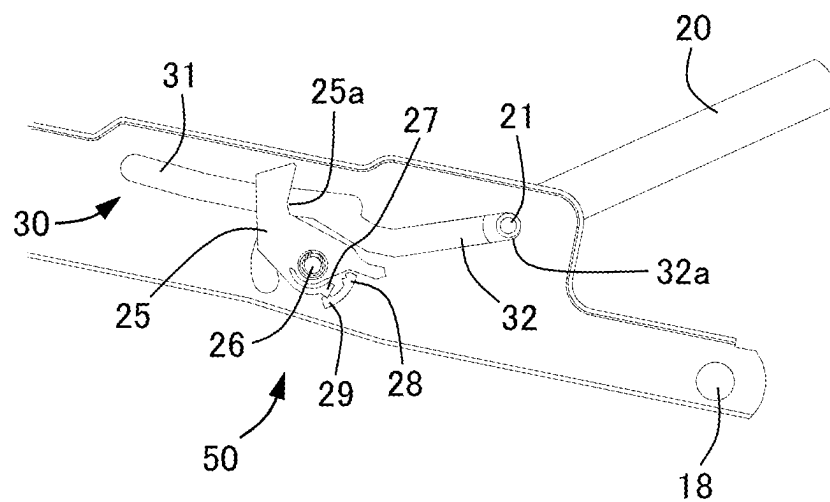


FIG.11B



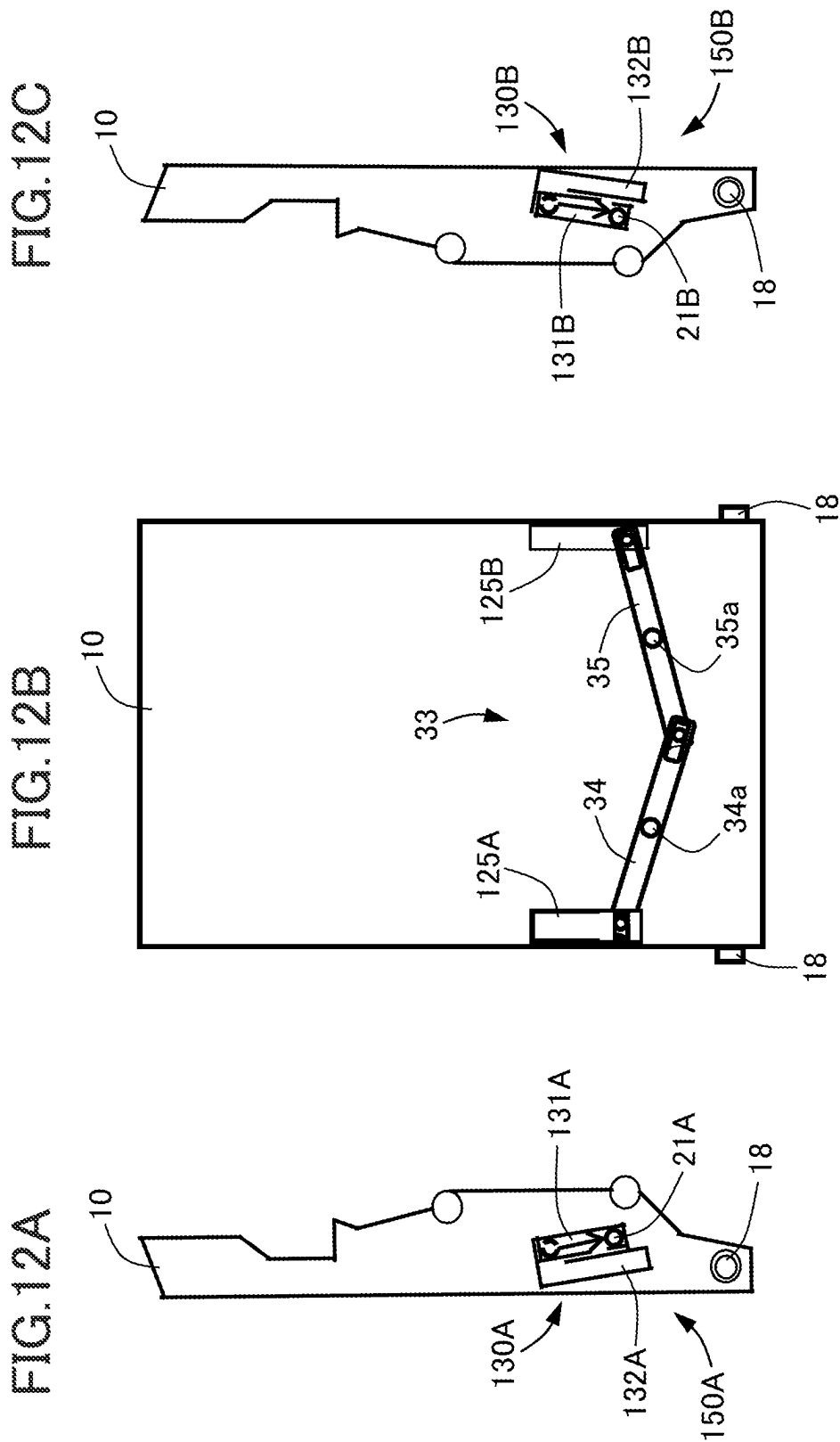


FIG.13A

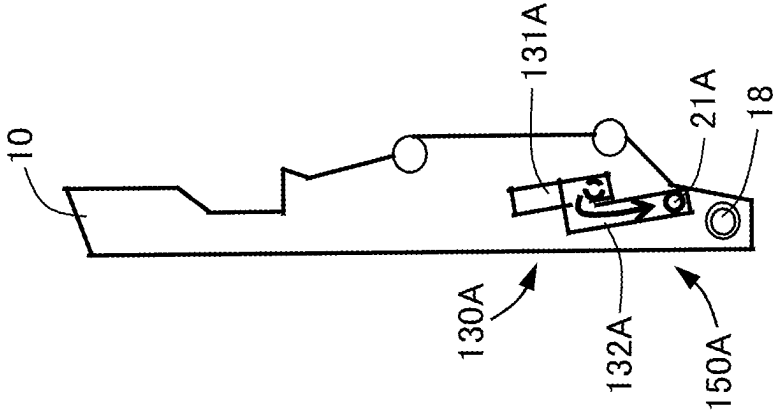


FIG.13B

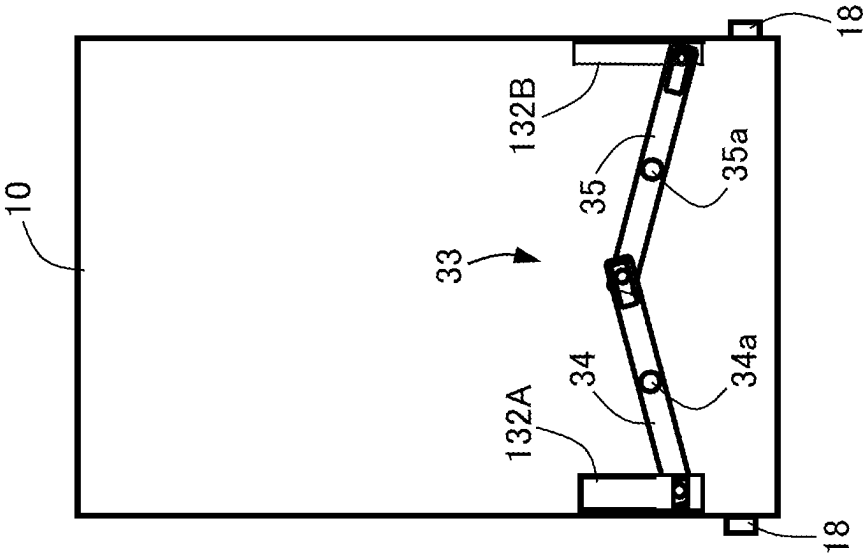


FIG.13C

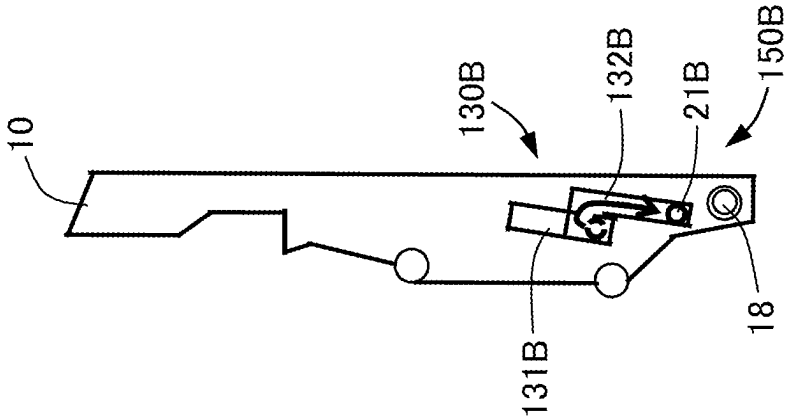


FIG.14A

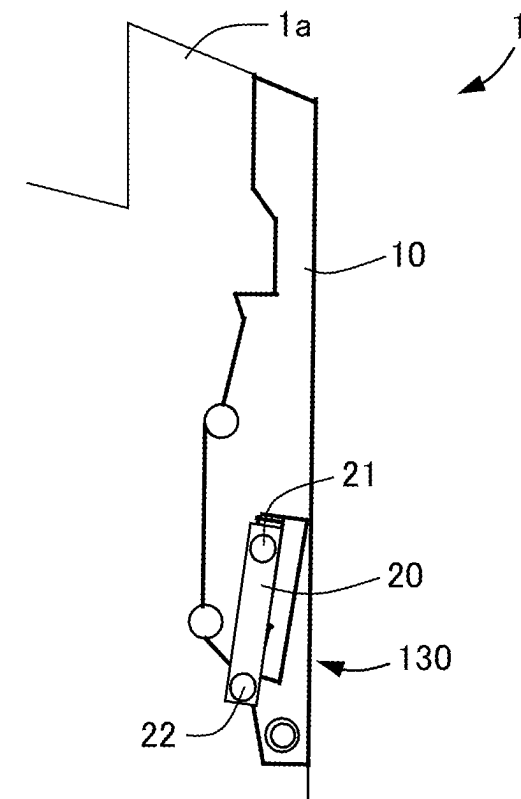


FIG.14B

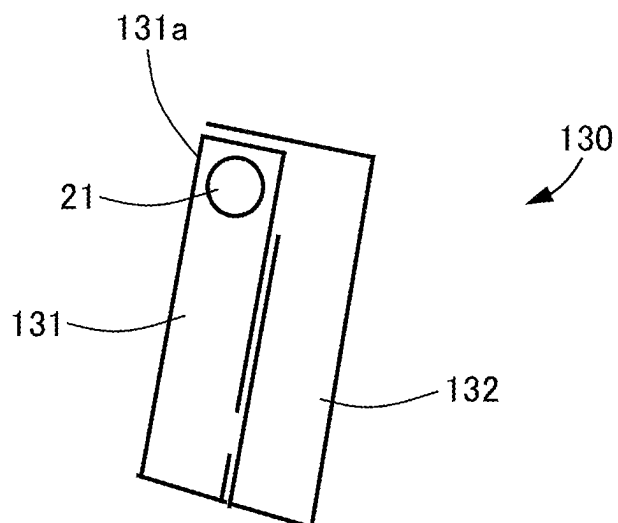


FIG.15A

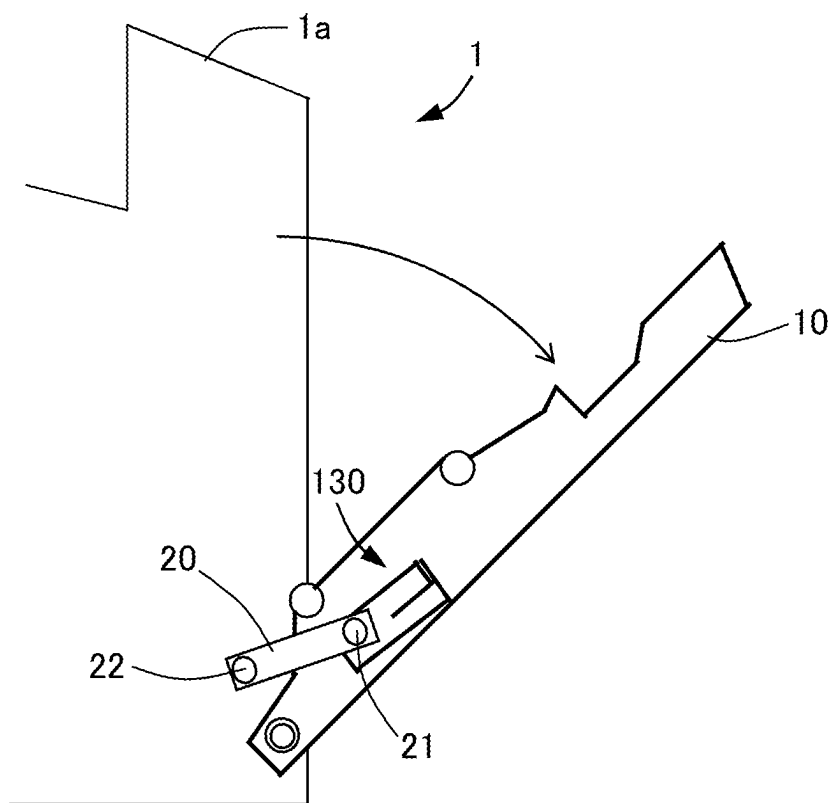


FIG.15B

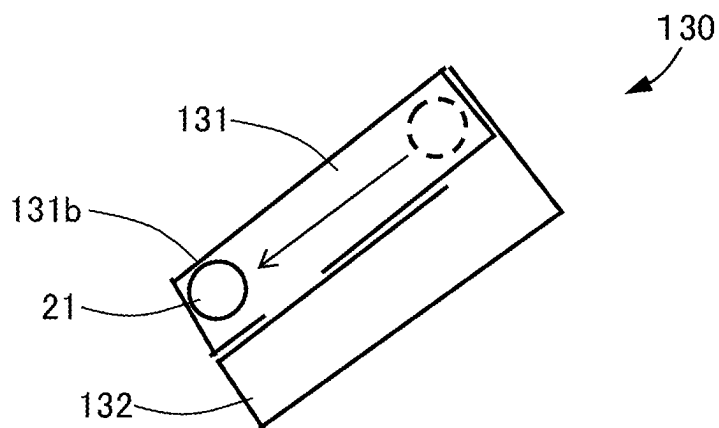


FIG.16A

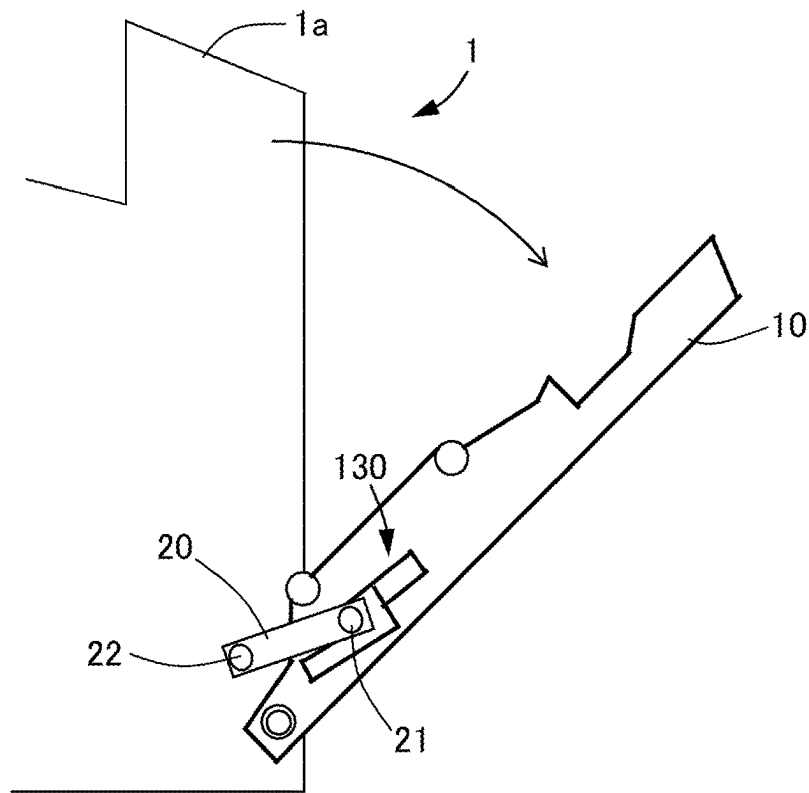


FIG.16B

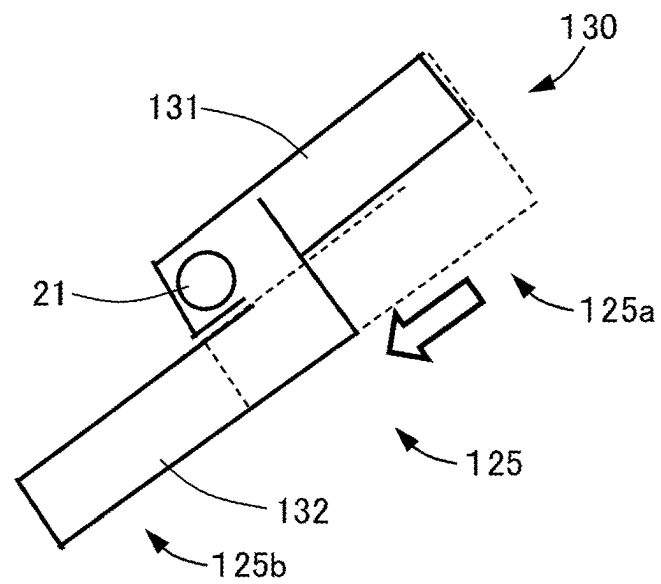


FIG.17A

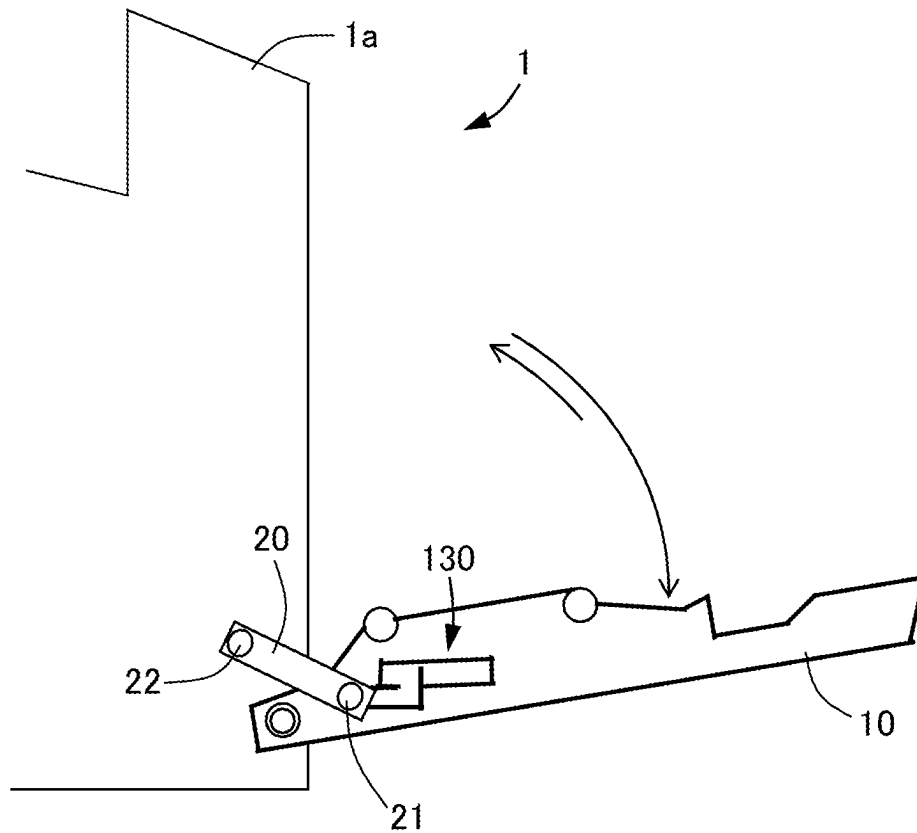


FIG.17B

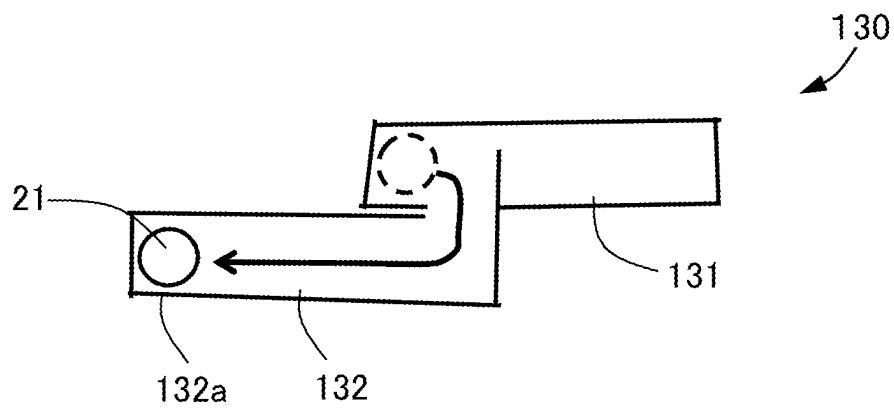


FIG.18A

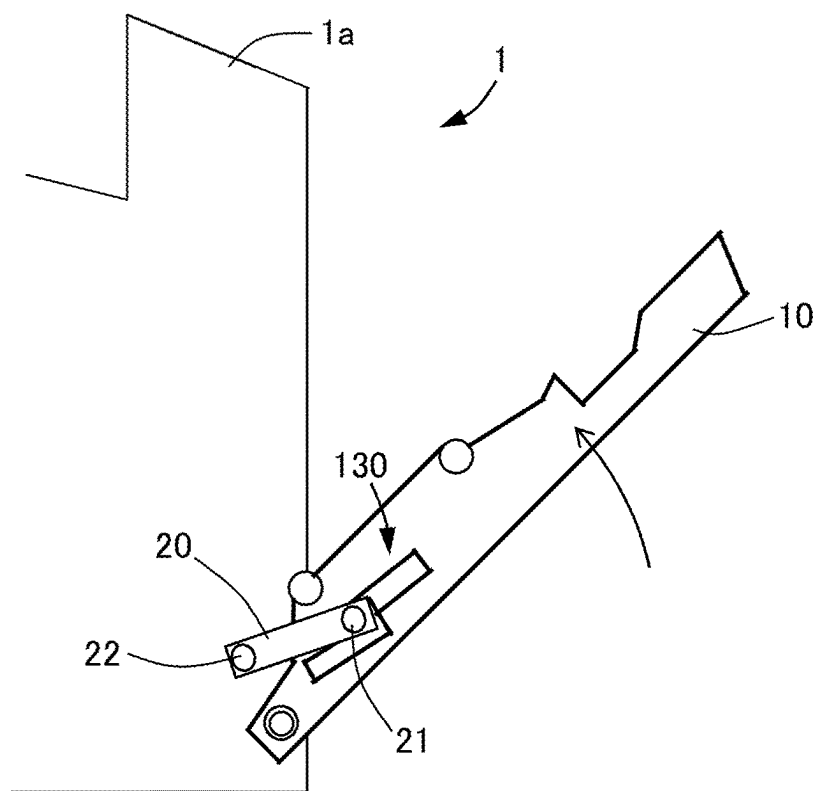


FIG.18B

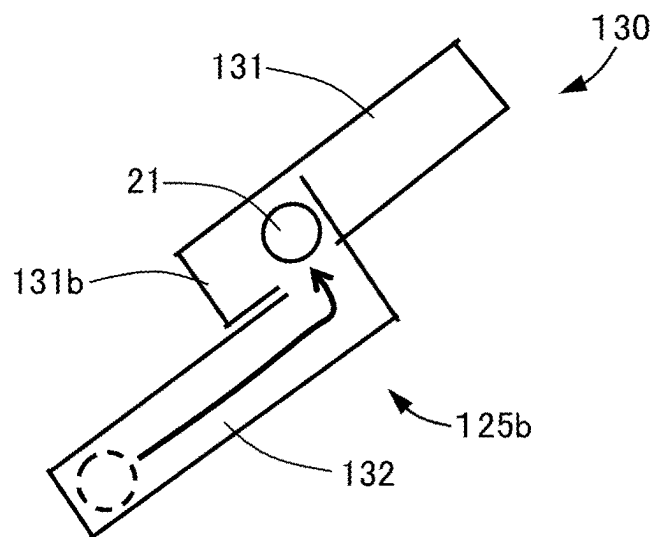


FIG.19A

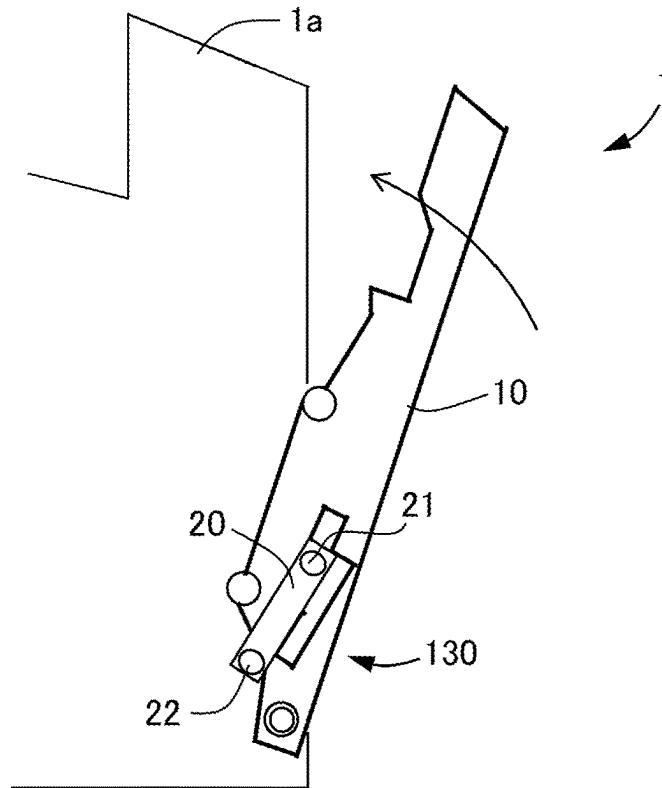


FIG.19B

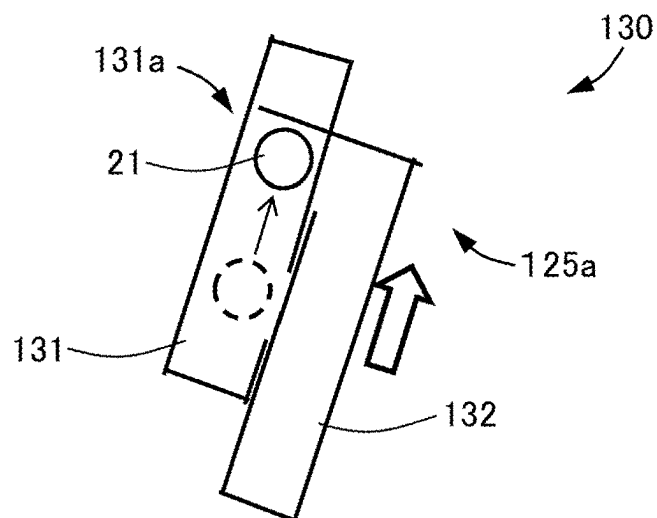


FIG.20

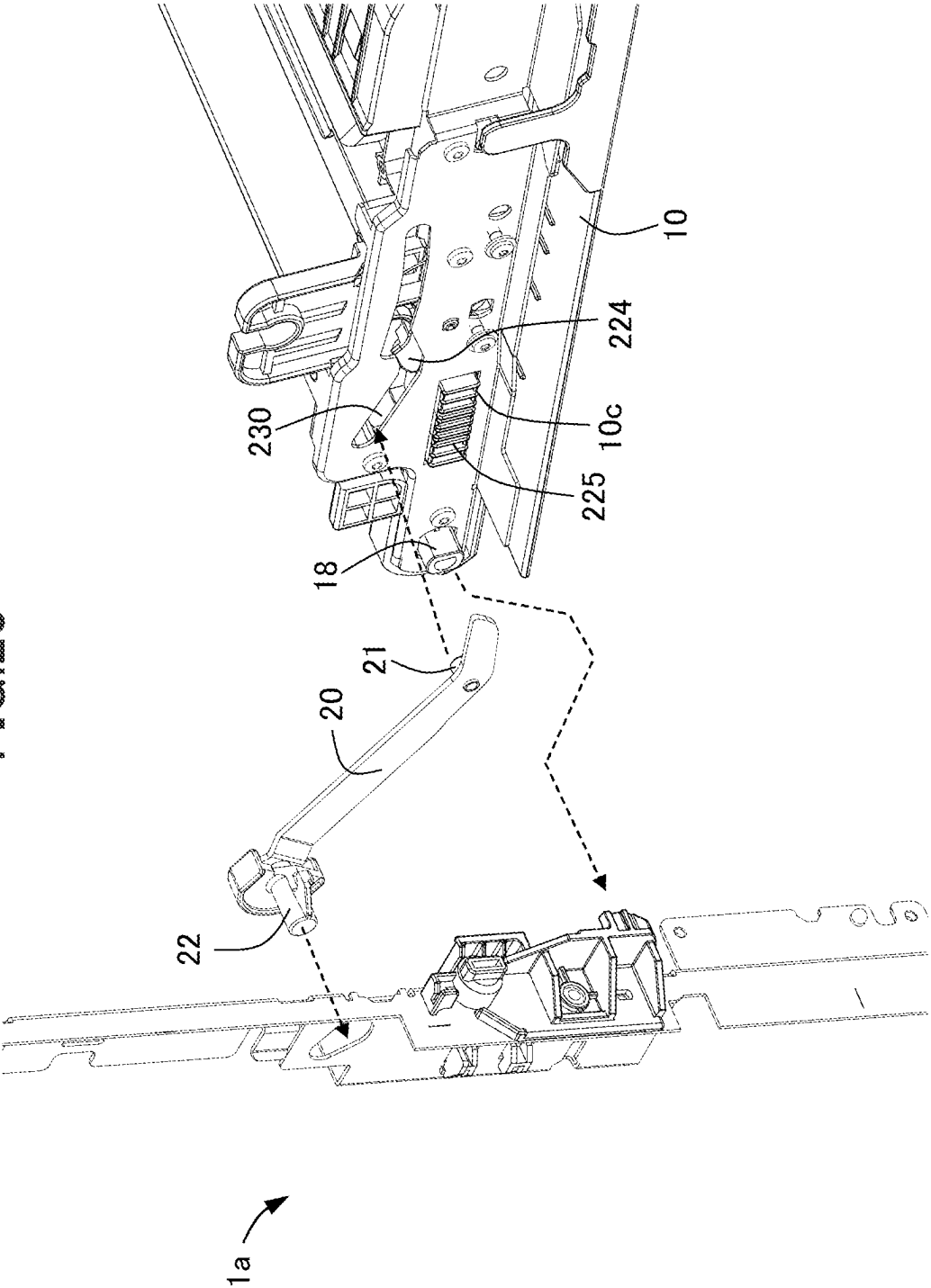


FIG.21

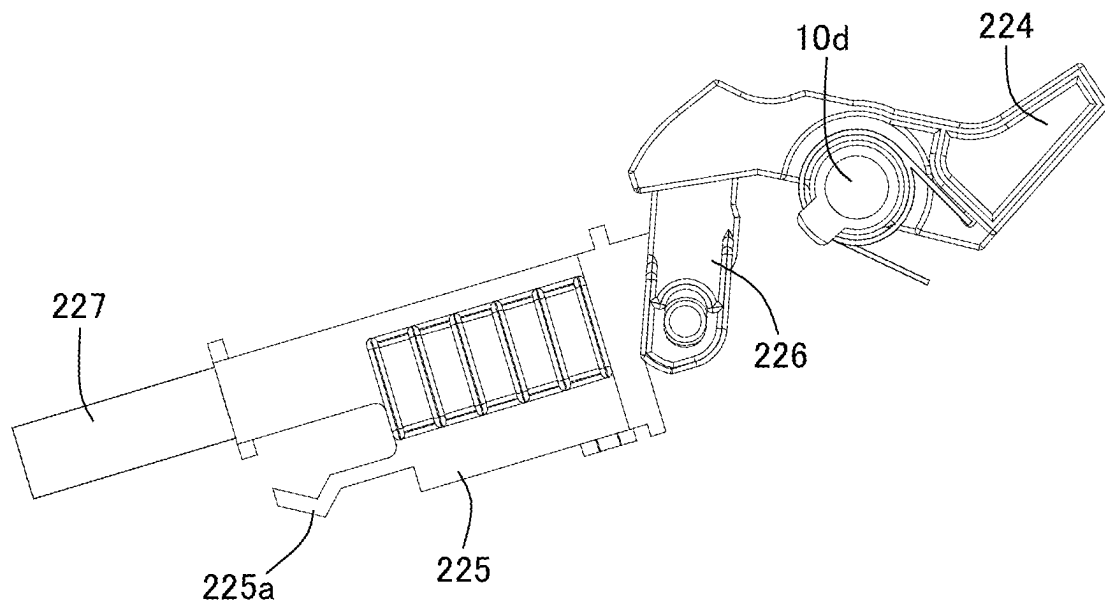


FIG.22

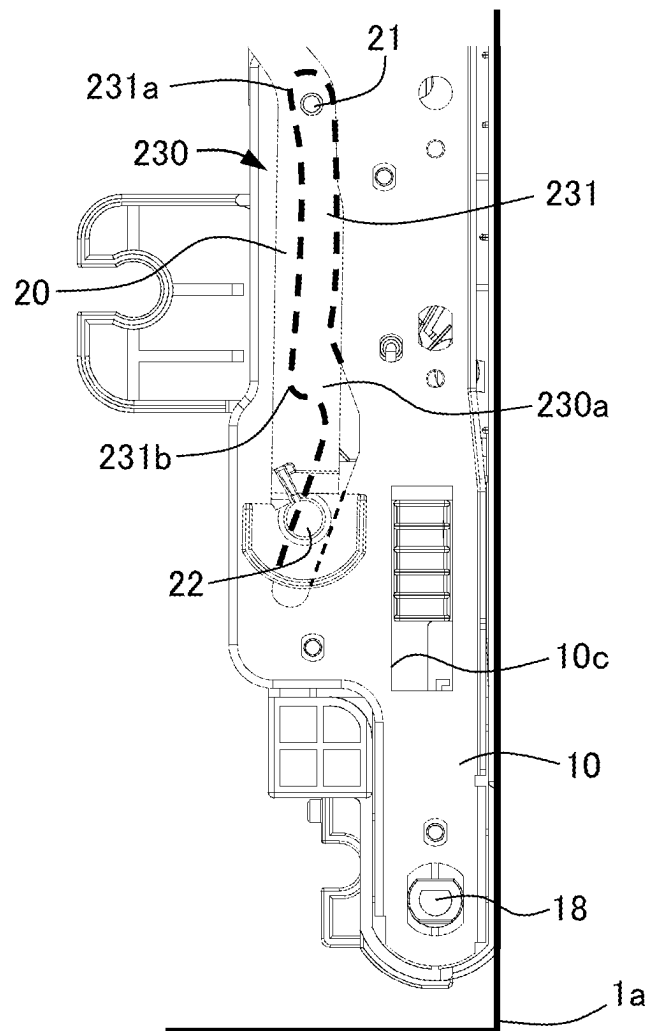


FIG.23

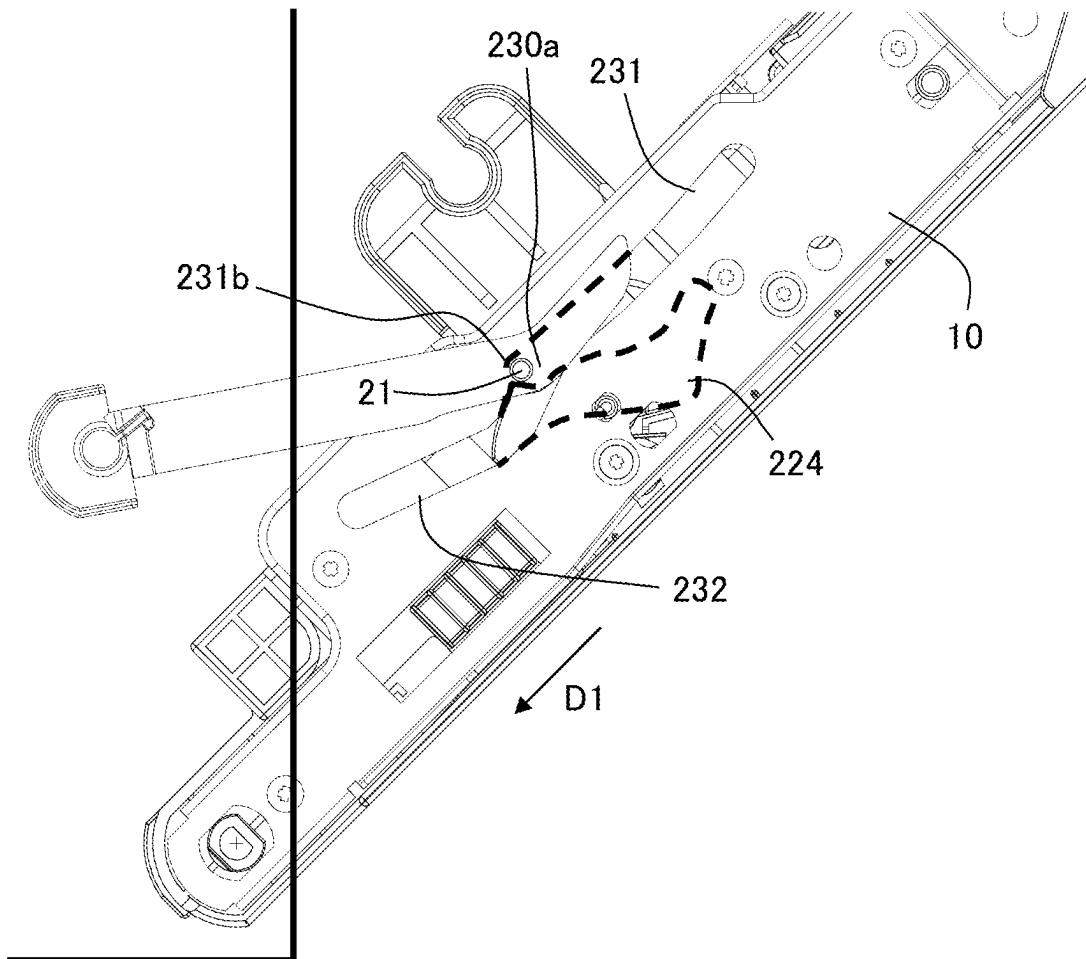
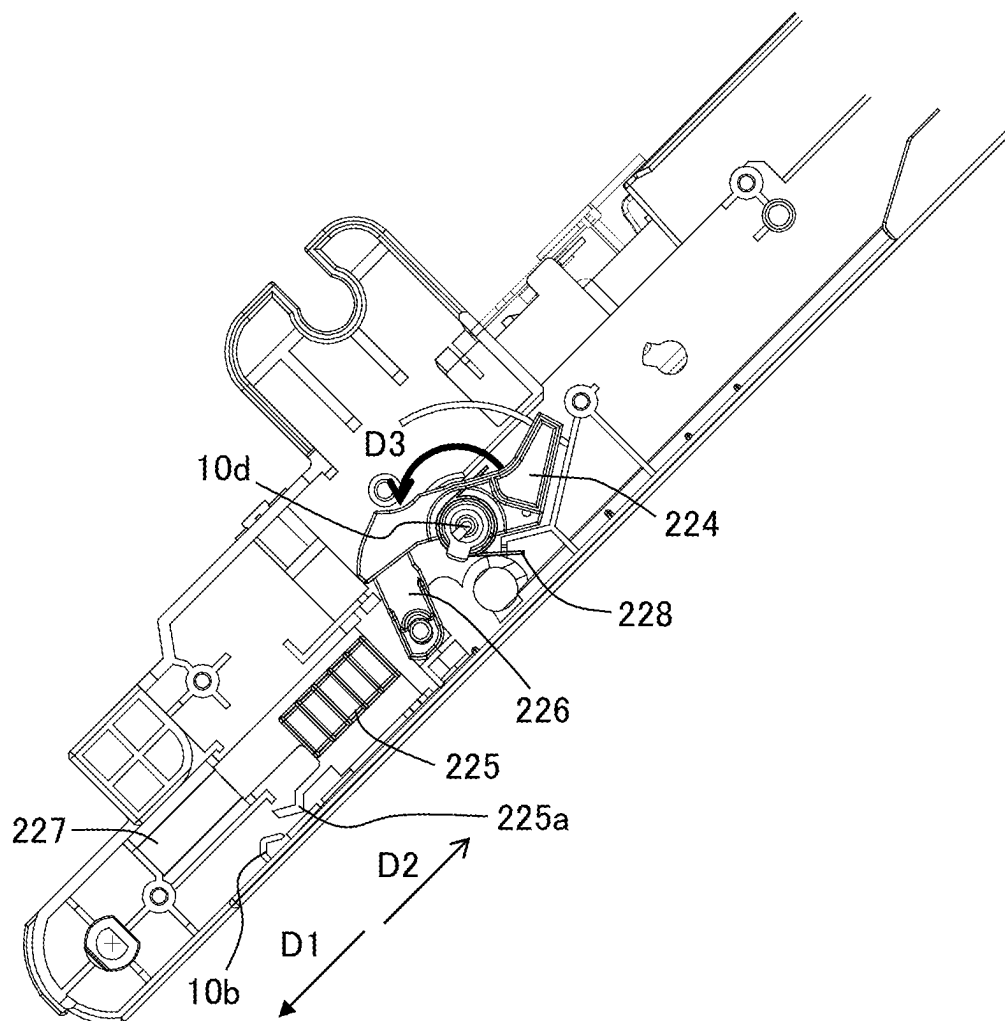


FIG.24



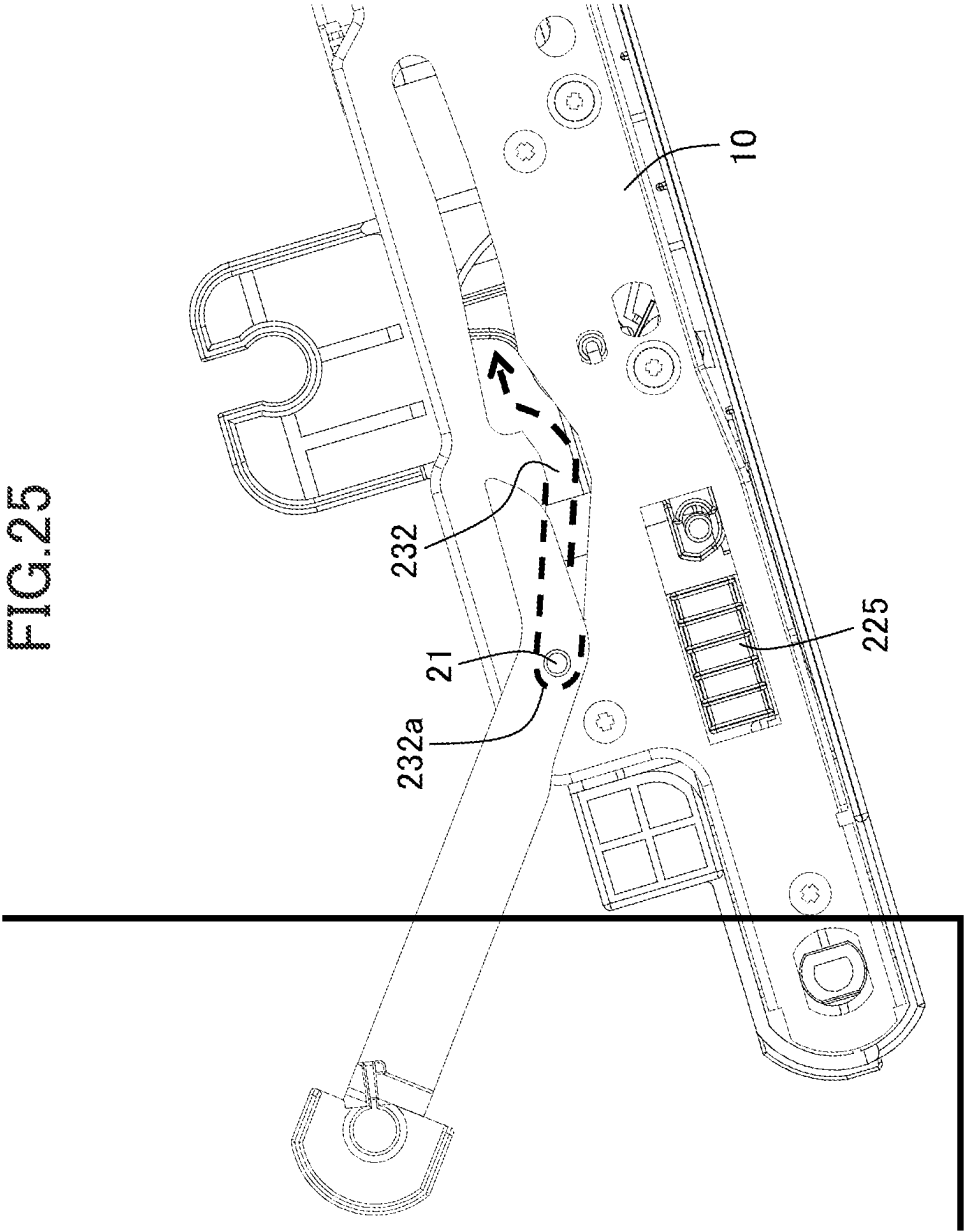


FIG.26

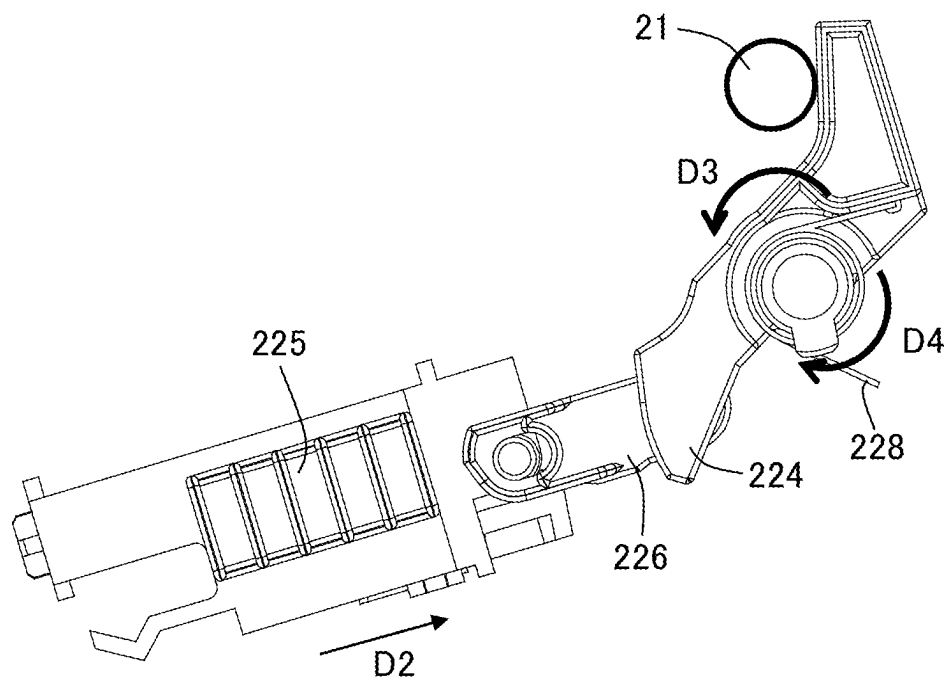


FIG.27

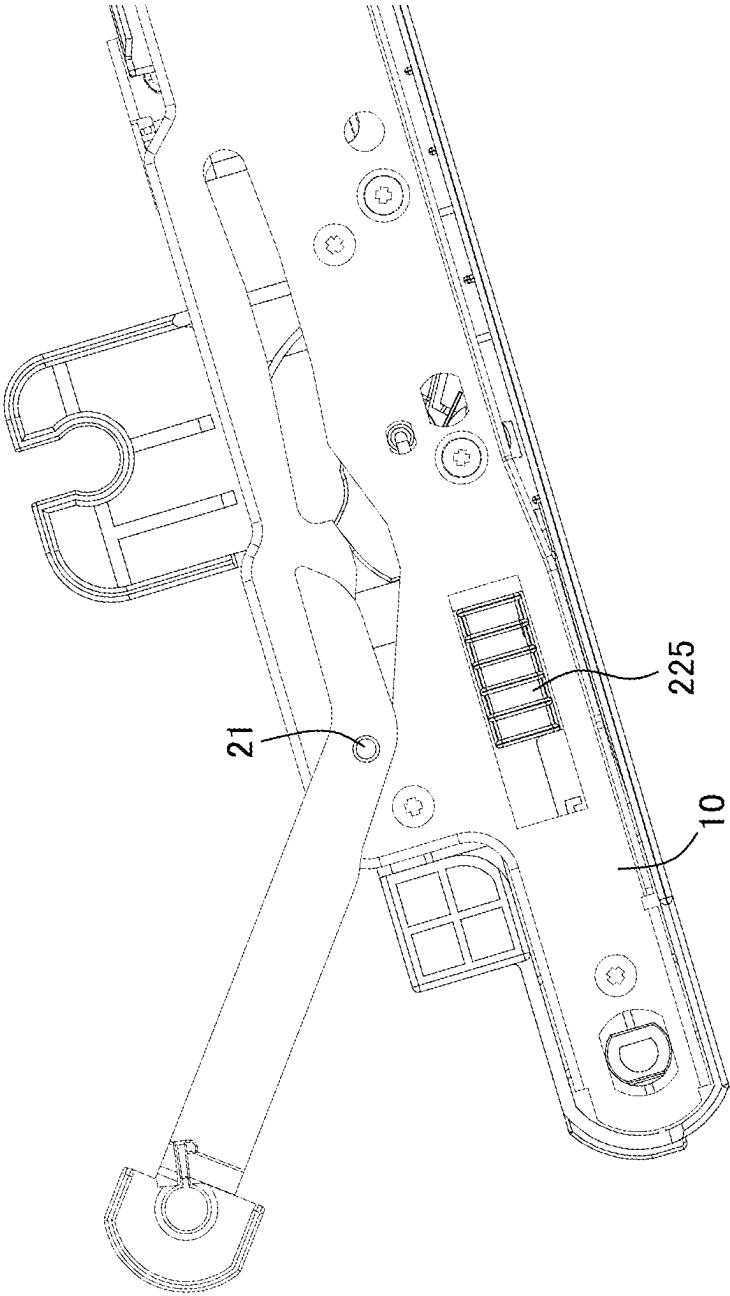


FIG.28

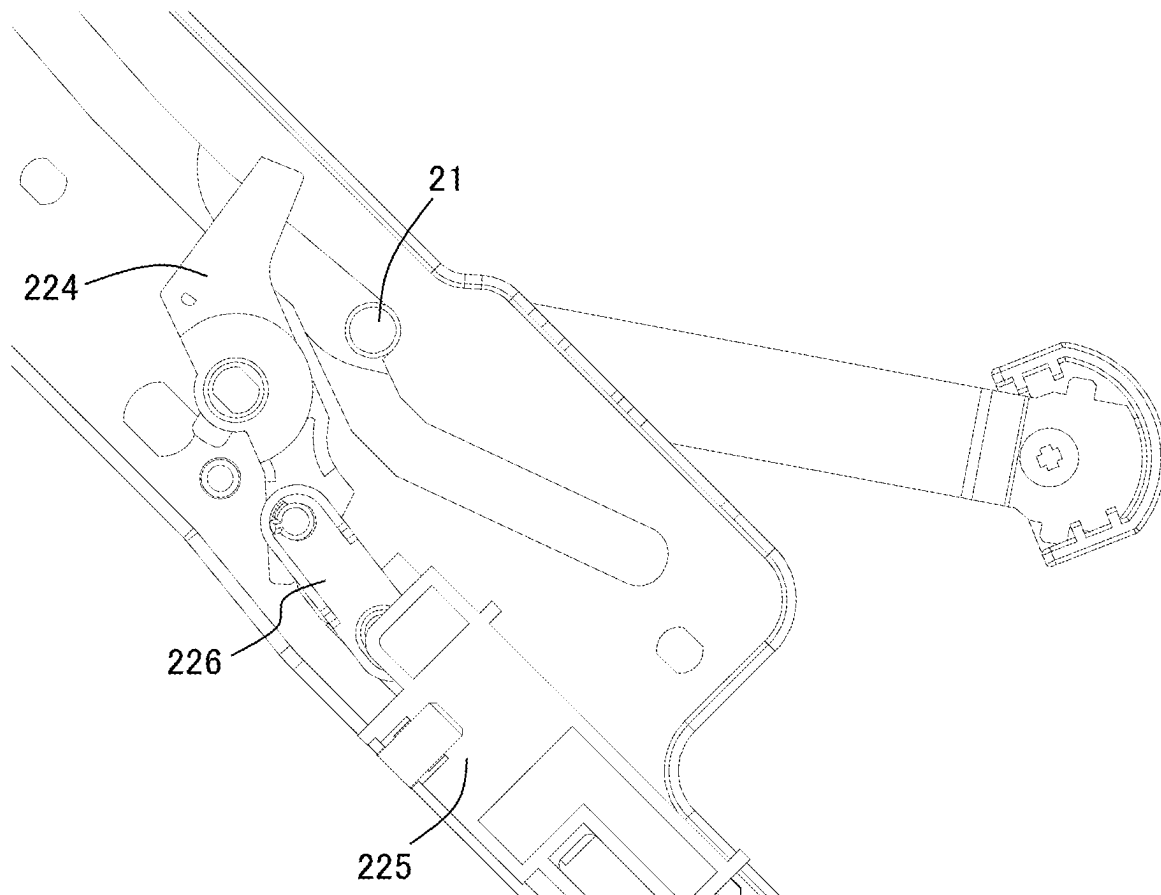


FIG.29A

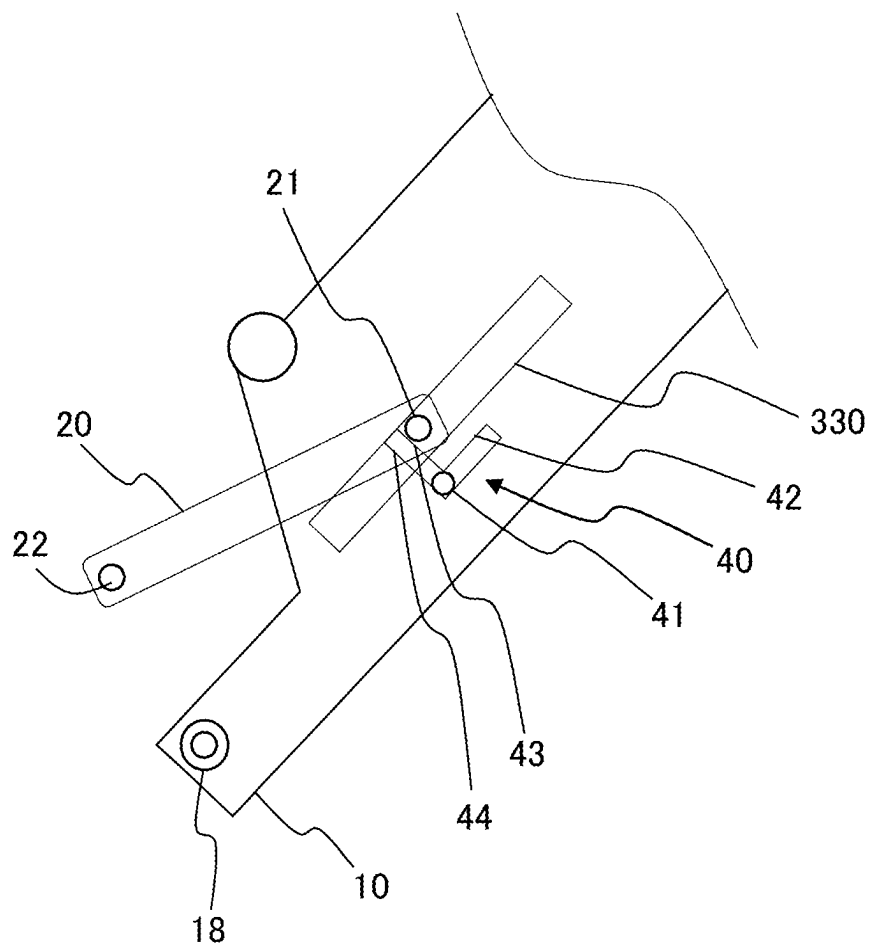


FIG.29B

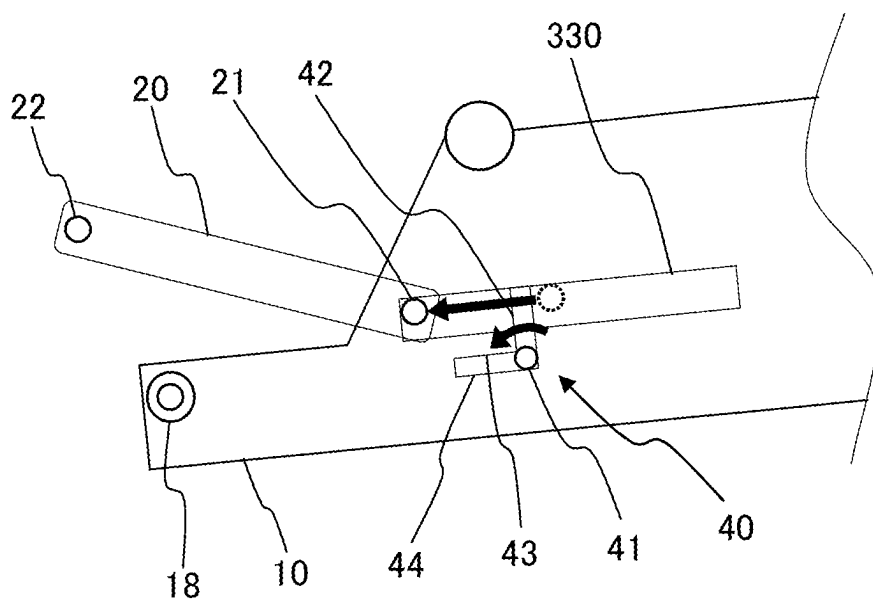


FIG.30A

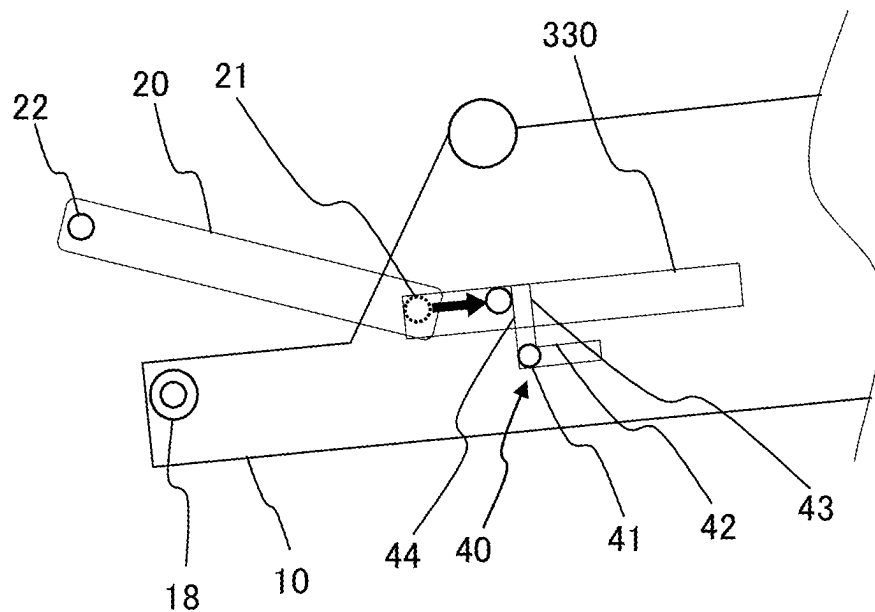


FIG.30B

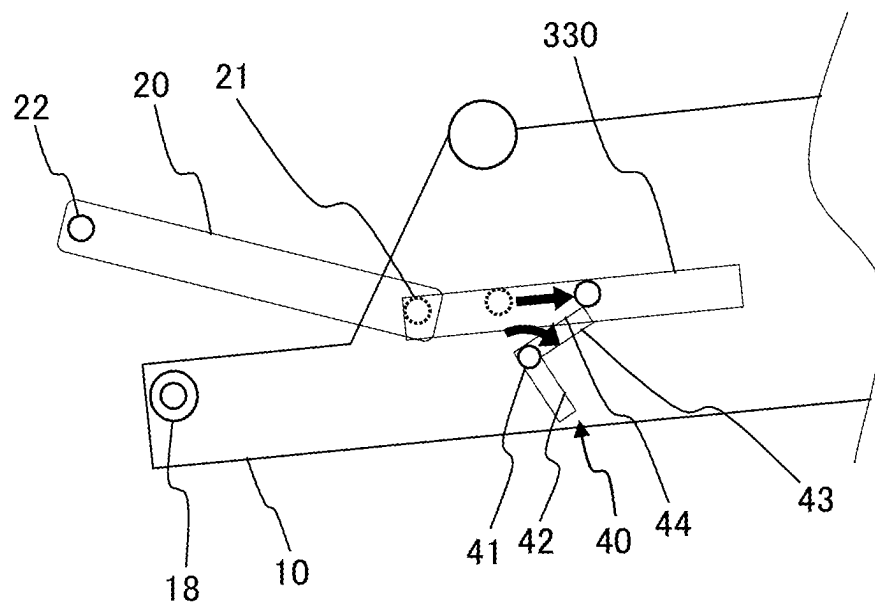


FIG.31

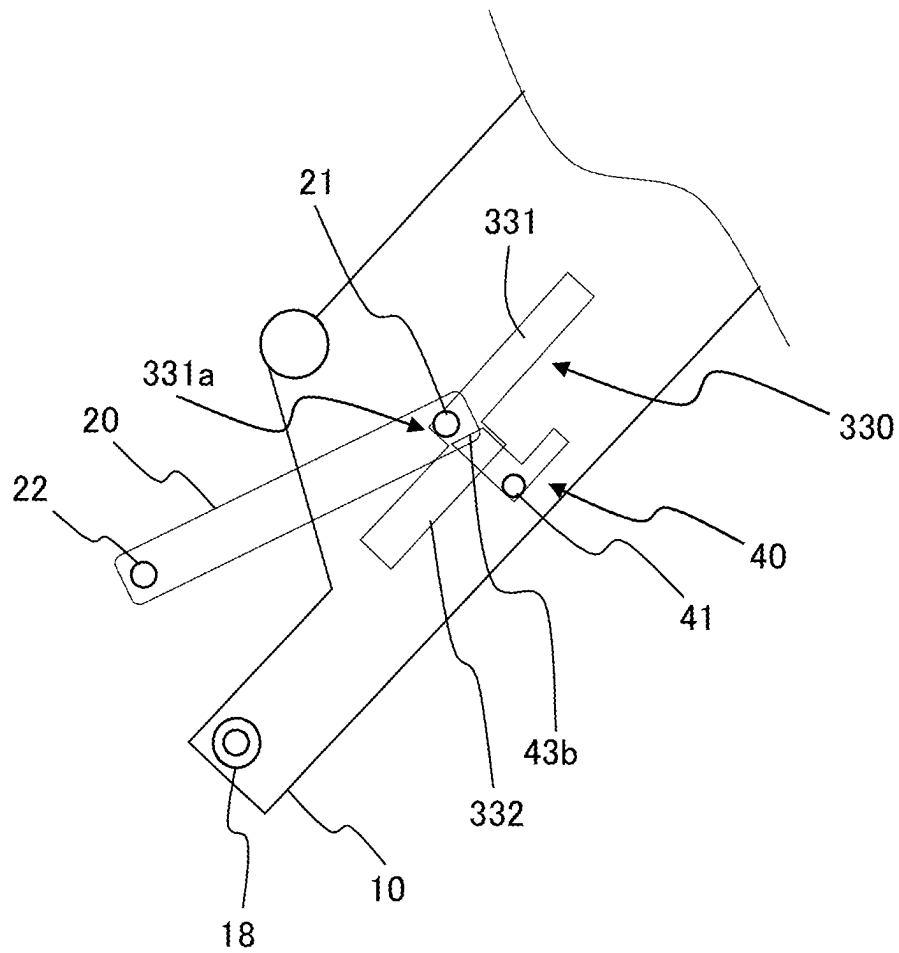


FIG.32A

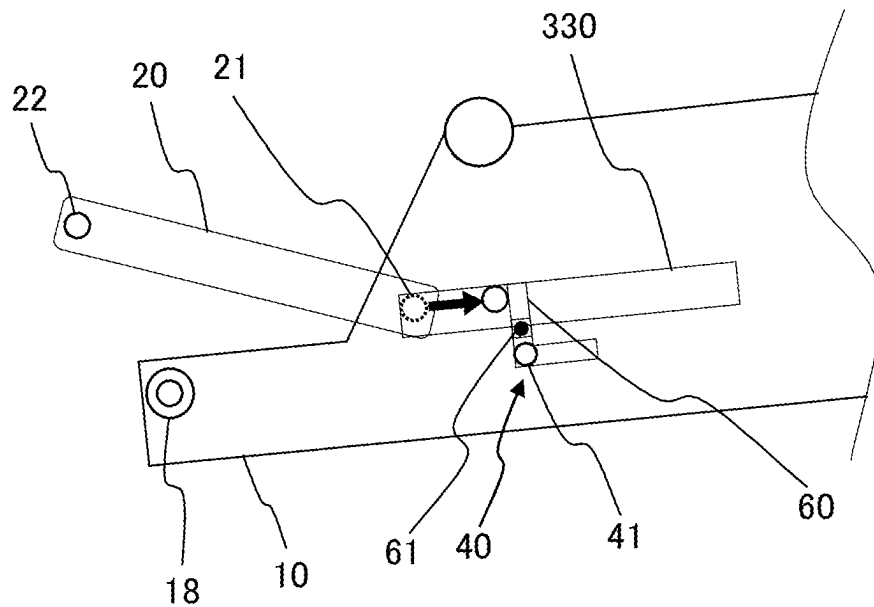
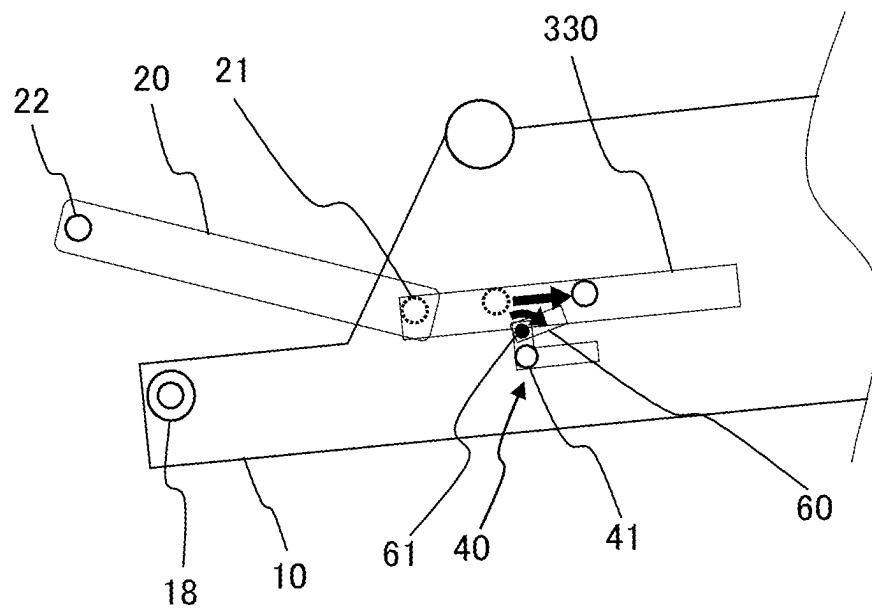


FIG.32B



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IMAGE FORMING APPARATUS**BACKGROUND OF THE INVENTION****Field of the Invention**

The present invention relates to image forming apparatuses adopting an electrophotographic system, such as a facsimile, a printer, or a multifunction machine having such functions.

Description of the Related Art

Hitherto, in image forming apparatuses, a configuration in which a conveyance path is formed as an openable/closable cover is adopted to cope with jamming of sheets, hereinafter referred to as sheet jamming, in the conveyance path through which sheets are conveyed. In addition to removing the sheets that caused sheet jamming, the cover is opened and closed when attaching, detaching or replacing components and units, such as an intermediate transfer belt, a fixing unit, and a conveyance roller, that are arranged along the conveyance path. A configuration in which an opening angle of a cover is set to two stages, which are a small angle and a large angle, has been proposed (refer to Japanese Patent Application Laid-Open Publication No. 2014-21135). For example, since the opening angle for removing jammed sheets may be set small, the cover is opened to a small angle so as to reduce the processing space required for removing jammed sheets. Further, by switching the opening angle to a larger angle, attaching and detaching of components can be performed conveniently.

The above-mentioned configuration proposes a two-action system as a unit for switching the opening angles of the cover, wherein the cover is lifted to a point where the opening angles are switched, and while the cover is held in this state, a member supporting the cover is operated to switch angles. Actually, a guide groove that guides a guide shaft at a tip of a support arm disposed swingably on a casing is formed on the cover, and a small angle path, a large angle path, and a branching point are formed in the guide groove. Thereby, the operation for switching from the small angle to the large angle may be performed by an operation of lifting the support arm and switching the guide shaft from the small angle path to the large angle path. According to this configuration, when switching the opening angle, the support arm must be held by the user, such that the cover will not be opened with force unintentionally from the small angle to the large angle, and damaging of components caused by impact may be prevented.

However, according to the image forming apparatus disclosed in Japanese Patent Application Laid-Open Publication No. 2014-21135, when opening the cover from the small angle to the large angle, there is a need to lift the support arm while guiding the guide shaft from the small angle path to the large angle path. According to this configuration, the load of the cover is received by the support arm of the cover, such that a large load is applied to the support arm, and this state is even more outstanding in a printer corresponding to large-size sheets or a high-speed apparatus in which the cover has a large size and has various functions. Therefore, in order to open the cover from the small angle to the large angle, such a heavy support arm must be lifted while performing the cover opening operation, such that a large operating force is required and the operability is deteriorated. Since the operability is not good, the operation of opening the cover to a large opening angle

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could only be performed by operators having special skills. Therefore, simplification of the operability was desired to allow general users to operate the cover.

The present technique provides an image forming apparatus capable of simplifying the operability of opening the cover in two stages.

SUMMARY OF THE INVENTION

According to a first aspect of the present invention, an image forming apparatus includes an image forming unit configured to form an image on a sheet, a casing accommodating the image forming unit, a door configured to be opened and closed with respect to the casing, and configured to be moved to a first opened position and a second opened position, the first opened position being a position where the door is inclined by a first angle with respect to a closed position, where the casing is closed and a sheet conveyance path is formed, to expose the sheet conveyance path, the second opened position being a position where the door is inclined by a second angle that is greater than the first angle with respect to the closed position to expose the sheet conveyance path, a connecting member configured to be interlocked with an opening and closing operation of the door, and connected to the casing and the door, a guide member provided on the door, the guide member including a first guide configured to guide the connecting member such that the door is positioned at the closed position and the first opened position respectively, and a second guide branched from the first guide and configured to guide the connecting member such that the door is positioned at the second opened position, and an operating portion configured to move, through an operation by a user in a state where the door is at the first opened position, from a regulating position where the connecting member is regulated from moving from the first guide to the second guide, to an allowing position where the connecting member is allowed to move from the first guide to the second guide. In a state where the operating portion is at the regulating position, the door is configured to be retained at the first opened position by engagement of the connecting member and the first guide. In a state where the operating portion is at the allowing position, the door is configured to be retained at the second opened position by movement of the connecting member from the first guide to the second guide.

Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a cross-sectional view illustrating a general configuration of an image forming apparatus according to a first embodiment.

FIG. 2A is a front view of a cover in a state where the cover according to the first embodiment is positioned at a closed position.

FIG. 2B is a front view of a guide portion in a state where the cover according to the first embodiment is positioned at the closed position.

FIG. 3A is a front view of the cover in a state where the cover according to the first embodiment is positioned at a first opened position.

FIG. 3B is a front view of the guide portion in a state where the cover according to the first embodiment is positioned at the first opened position.

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FIG. 23 is a front view of a state where the cover according to the third embodiment is positioned at a first opened position.

FIG. 24 is a front view of an internal configuration in a state where the cover according to the third embodiment is positioned at the first opened position.

FIG. 25 is a front view of the cover according to the third embodiment positioned at a second opened position.

FIG. 26 is an explanatory view of a state where the cover according to the third embodiment is moved from the second opened position to the closed position.

FIG. 27 is an explanatory view illustrating a configuration for preventing erroneous operation of the cover according to the third embodiment.

FIG. 28 is an explanatory view in a state where the cover according to the third embodiment is moved from the first opened position to the second opened position.

FIG. 29A is a front view of a cover according to a fourth embodiment in a state where a fixing member is in a first state.

FIG. 29B is a front view of the cover according to a fourth embodiment in a state where the fixing member is in a second state.

FIG. 30A is a front view of the cover according to a fourth embodiment in a state where the fixing member has been returned from the second state to the first state.

FIG. 30B is a front view of the cover according to a fourth embodiment in a state where the fixing member has retreated from the first state.

FIG. 31 is a front view of a cover according to a modified example of the fourth embodiment in which a fixing member is in a first state.

FIG. 32A is a front view of a cover according to a fifth embodiment in which a fixing member has been returned from a second state to a first state.

FIG. 32B is a front view of the cover according to the fifth embodiment in a state where a movable member of a fixing member has been rotated.

DESCRIPTION OF THE EMBODIMENTS

First Embodiment

A first embodiment of the present invention will be described in detail below with reference to FIGS. 1 to 11. In the present embodiment, a tandem-type full-color printer is described as an example of an image forming apparatus 1. However, the present invention is not limited to a tandem-type image forming apparatus 1, and it may be other types of image forming apparatuses, and it may even be applied not only to full-color printers but also to monochrome, mono-color, and inkjet printers.

Image Forming Apparatus

At first, a general configuration of the image forming apparatus 1 according to a first embodiment will be described with reference to FIG. 1. FIG. 1 is a schematic drawing of a color image forming apparatus using an electrophotographic system. The image forming apparatus 1 is an intermediate transfer tandem-type image forming apparatus in which four color image forming units of yellow (Y), magenta (M), cyan (C), and black (Bk) are aligned above the intermediate transfer belt 2. A direction vertical to the sheet surface in FIG. 1 is a front-rear direction of the apparatus.

The image forming apparatus 1 includes a casing 1a, and process cartridges Pa, Pb, Pc, and Pd stored in the casing 1a for forming toner images. The process cartridges Pa, Pb, Pc, and Pd are examples of an image forming portion and form

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an image on a sheet. The apparatus body includes the casing 1a and the process cartridges Pa, Pb, Pc, and Pd. The process cartridges Pa, Pb, Pc, and Pd use toner of different colors, which are yellow, magenta, cyan, and black, and they adopt similar configurations except for the different toner colors. Therefore, a process cartridge Pa of yellow toner is described below as a representative example.

In FIG. 1, a surface of a photosensitive member 6 that is driven to rotate in a clockwise direction is exposed and scanned by an exposing unit 7 driven based on an image information signal, and an electrostatic latent image is formed thereby. The electrostatic latent image is developed by the process cartridges Pa, Pb, Pc, and Pd as a toner image. Thereafter, predetermined pressurizing force and primary transfer bias are applied by a primary transfer apparatus T1, and a toner image is formed on the intermediate transfer belt 2. A primary transfer bias roller 5 and a transfer cleaning apparatus 4 are in contact with the intermediate transfer belt 2. Further, the intermediate transfer belt 2 is an endless belt that is stretched across a secondary transfer inner roller 9, a tension roller 3, and a secondary transfer upstream roller 8 that also function to transmit drive to the intermediate transfer belt 2, and that is driven to be conveyed in a counterclockwise direction in the drawing.

The primary transfer bias roller 5 forms a primary transfer nip by nipping the intermediate transfer belt 2 with the photosensitive member 6. An image forming process is performed at a timing at which a toner image is superposed on a color toner image that has been primarily transferred upstream on the intermediate transfer belt 2 by sequentially passing through the primary transfer nip, along with the conveyance of the intermediate transfer belt 2. Four color toner images are formed on the intermediate transfer belt 2 and conveyed to a secondary transfer unit T2.

A sheet S, i.e., storage medium, that has been stacked and stored in a sheet cassette 11 is fed one by one by a feed roller 12, and skewing of the sheet S is corrected by a registration roller 13 before the sheet S is transmitted at an appropriate timing to the secondary transfer unit T2. In the secondary transfer unit T2, a nip is formed by interposing the intermediate transfer belt 2 between the secondary transfer inner roller 9 and a secondary transfer outer roller 14, and the four-color toner image formed on the intermediate transfer belt 2 is secondarily transferred to the sheet S. The sheet S after secondary transfer is conveyed to a fixing unit 15, where the toner image on the sheet S is fixed, and the sheet S is discharged to an exterior of the apparatus by a sheet discharge roller pair 16 and sequentially stacked on a sheet discharge tray 17.

General Configuration of Opening/Closing Mechanism of Cover

Next, an opening/closing mechanism of a cover 10 will be described. An outline of the opening/closing mechanism for setting an angle, an orientation, and a position of the cover 10 will be described. FIG. 2A is an explanatory view of a closed position state, i.e., sheet conveyable state, or standby state, of the opening/closing mechanism according to the present embodiment, FIG. 3A is an explanatory view of a first opened position, i.e., jammed sheet removable position, of the opening/closing mechanism, and FIG. 4A is an explanatory view of a second opened position, i.e., unit replaceable position, of the opening/closing mechanism. FIGS. 2B, 3B, and 4B respectively illustrate a position of a slide shaft 21 in a guide portion 30 corresponding to FIGS. 2A, 3A, and 4A.

Jamming of sheets S may occur within the conveyance path at a right side of the image forming apparatus 1, such

that the cover 10 that may be used to open and close this part of the apparatus is disposed on the casing 1a. The cover rotation shaft 18 is provided on the cover 10, and by having the image forming apparatus 1 engage with the cover rotation shaft 18, the cover 10 is supported rotatably about a horizontal axis in a front-rear direction. That is, according to the present embodiment, the cover 10 constitutes a right side wall portion of the casing 1a in the closed position, and it is disposed rotatably about the cover rotation shaft 18 in the horizontal direction at a lower part of the cover 10. The cover 10 is designed to have its angle, position, and orientation determined by a support arm 20, and the cover 10 moves arbitrarily among three positions, which are a closed position, a first opened position, and a second opened position. The closed position is a position at which the cover 10 is stored in the image forming apparatus 1, which is referred to as a sheet conveyable state or a standby state. In the closed position, the cover 10 closes the casing 1a and forms a sheet conveyance path. The first opened position is a position at which the operator may remove jammed sheets, which is referred to as a jammed sheet removable position. The second opened position is a position at which the operator may replace units, which is referred to as a unit replaceable position. That is, the cover 10 is disposed in an openable and closable manner by rotating about the casing 1a, and it is movable between a first opened position that is inclined by a first angle with respect to a closed position where the casing 1a is closed to expose the sheet conveyance path, and a second opened position that is inclined by a second angle greater than the first angle with respect to the closed position to expose the sheet conveyance path.

The support arm 20 has the slide shaft 21 disposed on a first end portion so as to connect the support arm 20 and the cover 10 and is rotatably supported at a second end portion by a rotation shaft 22 to the casing 1a of the image forming apparatus 1. The cover 10 includes the guide portion 30 that guides the support arm 20. The guide portion 30 includes two paths, which are a first path 31, i.e., first area, and a second path 32, i.e., second area. That is, the guide portion 30 is an example of a guide member disposed on the cover 10, and includes the first path 31 which is an example of a first guide and the second path 32 which is an example of a second guide. The first path 31 guides the support arm 20 such that the cover 10 is positioned at the closed position and at the first opened position. The second path 32 is branched from the first path 31, and guides the support arm 20 such that the cover 10 is positioned at the second opened position.

An angle, i.e., position or orientation, of the support arm 20 is determined by the position of the slide shaft 21 on the guide portion 30, and simultaneously, an angle of the cover 10 is also determined. That is, the support arm 20 is an example of a support member that is disposed swingably on the casing 1a, that includes the slide shaft 21 serving as a guided portion guided by the guide portion 30, and that can support the cover 10 in an inclined state on the casing 1a. Further, the support arm is an example of a connecting member that is interlocked with the opening and closing operation of the cover 10, and that is connected to the casing 1a and the cover 10.

Closed Position, i.e., Sheet Conveyable State or Standby State

As illustrated in FIGS. 2A and 2B, when the slide shaft 21 of the support arm 20 is positioned on an end portion 31a of the first path 31, the support arm 20 is approximately parallel to the cover 10 in an approximately vertical direction, and the cover 10 is in a closed position stored in the image forming apparatus 1. In this state, by providing a lock

mechanism (not shown) to an appropriate position by which the cover 10 is locked to the image forming apparatus 1, the slide shaft 21 will not receive any load from the cover 10. In the closed position, which is the position taken during most of the period of use of the image forming apparatus 1, the slide shaft 21 will not receive load from the cover 10, such that it is not vulnerable to deterioration over time, such as creep deformation.

First Opened Position, i.e., Jammed Sheet Removable Position

As illustrated in FIGS. 3A and 3B, when the slide shaft 21 of the support arm 20 is positioned at an end portion 31b opposite to the closed position of the first path 31, the support arm 20 is in a balanced state by the rotation shaft 22 and the slide shaft 21 positioned at the end portion 31b. The cover 10 is positioned at the first opened position by the cover rotation shaft 18 and the slide shaft 21 positioned at the end portion 31b being balanced. The first opened position is required to provide an open space through which the operator can remove jammed sheets, and a first angle which is an opening angle thereof is preferably set, for example, to an angle of 45 degrees or greater and smaller than the angle of a second opened position. In the present embodiment, the first angle is set to 45 degrees. The first angle is not limited to 45 degrees or greater, and it can be set within the range of 30 to 60 degrees, for example.

Second Opened Position, i.e., Unit Replaceable Position

As illustrated in FIGS. 4A and 4B, when the slide shaft 21 of the support arm 20 is positioned at an end portion 32a of the second path 32, the support arm 20 is in a balanced state by the rotation shaft 22 and the slide shaft 21 positioned at the end portion 32a. The cover 10 is positioned at the second opened position by the balance between the cover rotation shaft 18 and the slide shaft 21 positioned at the end portion 32a. The second opened position is required to provide an open space through which the operator can replace units of the image forming apparatus 1 and parts of the cover 10, and a second angle which is an opening angle thereof is preferably set to an angle of 70 degrees or greater. In the present embodiment, the second angle is set to 75 degrees. The second angle is not limited to 70 degrees or greater, and it may be set within the range of 60 to 90 degrees, and may even be greater than 90 degrees.

Path Switching Mechanism

Next, a path switching mechanism 50 that switches the first path 31 and the second path 32 of the guide portion 30 will be described. According to the present embodiment, path switching mechanisms 50 are provided on both ends in the front and rear directions of the cover 10. However, since the configurations thereof are similar, only one of the path switching mechanisms 50 will be described. The path switching mechanism 50 may also be provided on only a front end portion of the cover 10.

FIGS. 5A to 8B are explanatory views showing a first path opened state of the path switching mechanism 50 according to the present embodiment. The second path 32 is formed to branch from a side surface of the first path 31. A path switching member 25 for switching paths between the first path 31 and the second path 32 is arranged at the branching point. The path switching member 25 is arranged rotatably about a rotation shaft 26 disposed on the cover 10, and it is engaged rotatably by an elastic engagement pawl 10a disposed on the cover 10. That is, the path switching mechanism 50 is an example of a rotating and switching portion, and the cover 10 may be switched between a first state in which the cover 10 is rotatable between the closed position

and the first angle and a second state in which the cover 10 is rotatable at least between the first angle and the second angle.

The path switching mechanism 50 includes the guide portion 30, the support arm 20, and the path switching member 25. The guide portion 30 includes the first path 31 and the second path 32. The first path 31 enables the cover 10 to rotate between the closed position and the first angle by guiding the slide shaft 21 therein. The second path 32 enables the cover 10 to rotate from an angle exceeding the first angle to the second angle by guiding the slide shaft 21 therein.

The path switching member 25 is an example of a regulation portion capable of regulating movement of the slide shaft 21, and includes a first path blocking portion 25a that blocks the first path 31 and a second path blocking portion 25b that blocks the second path 32. In a state where the first path blocking portion 25a protrudes into the path of the guide portion 30, the first path 31 is blocked and the second path 32 is opened, such that the slide shaft 21 is enabled to move in the second path 32. Meanwhile, in a state where the second path blocking portion 25b protrudes into the path, the first path 31 is opened, such that the slide shaft 21 is enabled to move in the first path 31. That is, according to the path switching member 25, in a state where the slide shaft 21 is positioned on the first path 31, by having the second path blocking portion 25b protrude into the path and restrict entry to the second path 32, the path switching mechanism 50 is set to a first state (refer to FIGS. 5A to 6B). Moreover, according to the path switching member 25, in a state where the slide shaft 21 is positioned at the end portion 31b of the first path 31, by having the first path blocking portion 25a protrude into the path to communicate the first path 31 and the second path 32, the path switching mechanism 50 is set to a second state (refer to FIGS. 7A to 8B). In a state where the path switching mechanism 50 is in the second state, the first path blocking portion 25a serves as an example of a contact portion that intersects a trajectory of the slide shaft 21 in the first path 31 of the guide portion 30. As described, the support arm 20 may be switched between a first state in which the cover 10 is rotatable between the closed position and the first opened position and a second state in which the cover 10 is rotatable at least between the first opened state and the second opened state. A holding portion 25c described later is used to switch the support arm 20 between the first state and the second state.

In order to enable the path switching member 25 to switch paths without fail, a mechanism to hold the position stably at each of a first path opening position where the first path 31 is opened and a second path opening position where the second path 32 is opened is required. According to the present embodiment, at first, in order to stabilize the position at the first path opening position, an engagement pawl 27 is provided on the path switching member 25, which is engaged with an engagement portion 28 of the cover 10 in a state where the path switching member 25 is positioned at the first path opening position. The engagement portion 28 is formed to protrude from an inner side of the cover 10. That is, the engagement portion 28 is an example of a retaining portion that retains the path switching member 25 such that the path switching mechanism 50 is maintained at the first state. Further, the path switching member 25 is released from retention by the engagement portion 28 through the operation of a holding portion 25c. A pawl member that deforms elastically is adopted as the engagement pawl 27 of the present embodiment considering costs, but the present technique is not limited thereto, and a

configuration may be adopted in which a member that is different from the path switching member 25 and that is moved in sliding motion by an elastic spring member may be provided, as long as it has repeatability. Simultaneously, in order to stabilize the position at the second path opening position, the engagement pawl 27 is engaged with an engagement portion 29 of the cover 10 in a state where the path switching member 25 is positioned at the second path opening position (refer to FIG. 7B).

The holding portion 25c that is used by the operator when performing a path switching operation is provided on the path switching member 25. That is, the holding portion 25c is one example of an operating portion for switching the path switching mechanism 50 between the first state and the second state. In a state where the cover 10 is positioned at the first opened position, the holding portion 25c is operated by the user to be moved from a regulating position in which the support arm 20 is regulated from moving from the first path 31 to the second path 32 to an allowing position in which the support arm 20 is allowed to move from the first path 31 to the second path 32. In the present embodiment, the holding portion 25c may be moved mutually between the regulating position and the allowing position. The operator operates the holding portion 25c to open the second path 32 when switching the cover 10 from the first opened position to the second opened position, and guides the path switching member 25 to the second path opening position. The switching operation is completed by engaging the engagement pawl 27 to the engagement portion 28 of the cover 10. The holding portion 25c is exposed toward an inner side from an inner wall of the cover 10, and it is provided at a part other than the part being an external wall of the casing 1a at the cover 10 in the closed position. Therefore, it becomes possible to prevent the user from operating the holding portion 25c unintendedly when the cover 10 is closed.

Meanwhile, in a state where the first path 31 is opened, the operator rotates the cover 10 to the closed position without operating the holding portion 25c. Thereby, switching is completed by the slide shaft 21 coming into contact with the first path blocking portion 25a, guiding the path switching member 25 by pushing the same to the first path opening position, and engaging the engagement pawl 27 to the engagement portion 28 of the cover 10. By adopting the above-mentioned configuration, a switching mechanism for switching the paths of the slide shaft 21 can be realized. That is, in a state where the holding portion 25c is at the regulating position, the support arm 20 and the first path 31 are engaged, and which the cover 10 is retained at the first opened position. Further, in a state where the holding portion 25c is in the allowing position, the support arm 20 is moved from the first path 31 to the second path 32, and the cover 10 is retained at the second opened position.

Opening/Closing Operation of Cover

An operation by which the operator switches the opening/closing angle of the cover 10 and a movement of the mechanism will be described. FIGS. 9A and 9B are an explanatory view of the opening/closing mechanism in the closed position state and the path switching mechanism 50 according to the present embodiment. FIGS. 10A and 10B are explanatory views of the opening/closing mechanism in the first opened position and the path switching mechanism 50. FIGS. 11A and 11B are explanatory views of the opening/closing mechanism in the second opened position and the path switching mechanism 50.

From Closed Position to First Opened Position

In order to open the cover 10 from the closed position, i.e., stored position, to the first opened position, i.e., jammed

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sheet removable position, if a lock mechanism is provided to lock the cover 10 to the image forming apparatus 1, the lock may be unlocked. When the opening operation is started, the slide shaft 21 of the support arm 20 slides along the first path 31 from an end portion 31a toward the end portion 31b. In this state, the path switching member 25 is at the first path opening position and the second path 32 is blocked, such that the slide shaft 21 is guided to the slide shaft 21 and reaches the end portion 31b, and the support arm 20 and the cover 10 are stopped in the balanced state. Thus, the opening operation to the first opened position, i.e., jammed sheet removable position, is completed.

From First Opened Position to Closed Position

When closing the cover 10 from the first opened position, i.e., jammed sheet removable position, to the closed position, i.e., stored position, the operator pushes the cover 10 toward the casing 1a of the image forming apparatus 1 using his/her hand. Thereby, the slide shaft 21 of the support arm 20 slides on the first path 31 from the end portion 31b toward the end portion 31a, and the lock mechanism is locked to the image forming apparatus 1, by which the movement is completed.

From First Opened Position to Second Opened Position

When opening the cover 10 from the first opened position, i.e., jammed sheet removable position, to the second opened position, i.e., unit replaceable position, at first, the path switching member 25 is moved to the second path opening position. The operator operates the holding portion 25c to move the same to the second path opening position, and the engagement pawl 27 is engaged from the engagement portion 28 to the engagement portion 29 of the cover 10. Thereby, the first path 31 is blocked, and instead, the second path 32 is opened. In this state, the cover 10 and the support arm 20 are still balanced and are still maintained at the first opened position. In this state, when the operator lifts the cover 10 slightly by which the slide shaft 21 is separated from the end portion 31b, the balanced state is cancelled and the slide shaft 21 is guided to the second path 32.

In this state, the cover 10 is not opened to the second opened position, i.e., unit replaceable position, immediately after switching the path switching member 25, and the balanced state is maintained. In other words, the path switching member 25 merely blocks the second path 32, and the load of the cover 10 is received by the end portion 31b provided on the first path 31. That is, the end portion 31b is an example of a receiving portion that receives the load of the cover 10 acting on the slide shaft 21 when the path switching mechanism 50 is in the first state and the cover 10 is positioned at the first opened position. Thereby, the only operating force for operating the path switching member 25 is the load that occurs when engaging the engagement pawl 27 from the engagement portion 29 to the engagement portion 28 of the cover 10. Thereby, the operating force exerted by the operator when performing operation can be reduced, and the life of the product can be extended. The slide shaft 21 guided to the second path 32 reaches the end portion 32a, and the support arm 20 and the cover 10 are stopped at the balanced state. Thereby, the opening operation to the second opened position, i.e., unit replaceable position, is completed.

From Second Opened Position to Closed Position

When closing the cover 10 from the second opened position, i.e., unit replaceable position, to the closed position, i.e., stored position, the operator pushes the cover 10 toward the casing 1a of the image forming apparatus 1 using his/her hand. Thereby, the slide shaft 21 of the support arm 20 slides on the second path 32 and the first path 31 from the

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end portion 32a toward the end portion 31a, and the lock mechanism is locked to the image forming apparatus 1, by which the operation is completed. In this state, the first path 31 is blocked by the path switching member 25, but the slide shaft 21 is in contact with the first path blocking portion 25a and pushes the path switching member 25 to guide the same toward the first path opening position. Then, switching is completed by engaging the engagement pawl 27 from the engagement portion 29 to the engagement portion 28 of the cover 10. That is, in a state where the slide shaft 21 is positioned at the second path 32, the cover 10 rotates toward the closed position, by which the slide shaft 21 abuts against and pushes the first path blocking portion 25a, such that the path switching mechanism 50 is switched from the second state to the first state.

Thereby, the operation from the closed position, i.e., stored position, to the first opened position, i.e., jammed sheet removable position, may be restarted. Thereby, when closing the cover 10 from the second opened position to the closed position, a configuration is adopted in which the path switching member 25 is returned to the first path opening position in a manner interlocked with the operation of the cover 10 without the operator operating the holding portion 25c. That is, the path switching mechanism 50 is switched from the first state to the second state by the operation of the holding portion 25c, while the path switching mechanism 50 is switched from the second state to the first state by rotating the cover 10 from the second angle to the closed position. Thereby, the possibility of the operator performing an erroneous operation, such as failure to operate the mechanism, may be reduced.

As described, according to the image forming apparatus 1 of the present embodiment, the path switching mechanism 50 includes the holding portion 25c that switches between the first state and the second state. Therefore, the cover 10 may be opened greatly from the first angle to the second angle by the operation of the holding portion 25c, such that the operability for opening the cover 10 in two stages can be simplified. That is, since the path switching operation of opening the cover 10 from the first stage to the second stage can only be realized by switching the holding portion 25c, a simultaneous operation of receiving the load of the cover 10 while operating the support arm to switch paths becomes unnecessary. Thereby, the operation for opening the cover to the second stage is simplified, such that the switching operation may be performed by anyone, instead of being limited to a dedicated operator.

According to the embodiment mentioned above, a case has been described in which the cover 10 is a cover disposed on a right side of the casing 1a, but the present technique is not limited thereto. For example, the cover may be disposed on the left side of the casing 1a, or the cover may be a front cover. According further to the present embodiment, a case has been described in which the cover rotation shaft 18 supporting the cover 10 is arranged in a horizontal direction, but the present technique is not limited thereto, and for example, the cover rotation shaft may be arranged in the vertical direction.

According to the embodiment described above, a case has been described in which the cover 10 is opened in two stages, but the present technique is not limited thereto, and a configuration can be adopted in which the cover 10 is opened in three or more stages. In that case, paths for the guide portion per stage and path switching members are provided.

According to the embodiment described above, a case has been described in which the cover 10 is a cover of the image

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forming apparatus **1** equipped with an image forming unit, but the present technique is not limited thereto, and the image forming apparatus may include a postprocessing apparatus such as a finisher, and the cover may be a casing of the postprocessing apparatus.

Second Embodiment

Next, a second embodiment of the present invention will be described in detail with reference to FIGS. **12A** to **19**. The present embodiment differs from the first embodiment in that path switching mechanisms **150** disposed at each end portion in the front-rear direction of the cover **10** are interlocked. Other configurations are similar to the first embodiment, such that the same reference numbers are assigned to corresponding components, and detailed descriptions thereof are omitted.

According to the present embodiment, as illustrated in FIGS. **12** and **13**, path switching mechanisms **150A** and **150B** are respectively arranged on end portions in the front-rear direction of the cover **10**. A configuration of an interlocking portion **33** that interlocks the path switching mechanisms **150A** and **150B** will be described later. The path switching mechanisms **150A** and **150B** adopt a symmetrical shape, and configurations thereof are similar, such that only one of the path switching mechanisms **150** will be described below with reference to FIGS. **14A** to **19B**.

General Configuration of Opening/Closing Mechanism of Cover

FIG. **14A** is an explanatory view of a closed position state, i.e., sheet conveyable state or standby state, of an opening/closing mechanism according to the present embodiment, FIG. **15A** is an explanatory view of a first opened position, i.e., jammed sheet removable position, of the opening/closing mechanism, and FIG. **17A** is an explanatory view of a second opened position, i.e., unit replaceable position, of the opening/closing mechanism. FIGS. **14B**, **15B**, and **17B** respectively illustrate the position of a slide shaft **21** in a guide portion **30** corresponding to FIGS. **14A**, **15A**, and **17A**. Closed Position, i.e., Sheet Conveyable State or Standby State

As illustrated in FIGS. **14A** and **14B**, in a state where the slide shaft **21** of a support arm **20** is positioned at an end portion **131a** of a first path **131**, the support arm **20** is approximately parallel to approximately a vertical direction of the cover **10**, and the cover **10** is at a closed position stored in the image forming apparatus **1**.

First Opened Position, i.e., Jammed Sheet Removable Position

As illustrated in FIGS. **15A** and **15B**, when the slide shaft **21** of the support arm **20** is positioned at an end portion **131b** opposite to the closed position of the first path **131**, the support arm **20** is in a balanced state by the rotation shaft **22** and the slide shaft **21** positioned at the end portion **131b**. The cover **10** is positioned at the first opened position by the cover rotation shaft **18** and the slide shaft **21** positioned at the end portion **131b** being balanced.

Second Opened Position, i.e., Unit Replaceable Position

As illustrated in FIGS. **17A** and **17B**, when the slide shaft **21** of the support arm **20** is positioned at an end portion **132a** of a second path **132**, the support arm **20** is in a balanced state by the rotation shaft **22** and the slide shaft **21** positioned at the end portion **132a**. The cover **10** is positioned at the second opened position by the cover rotation shaft **18** and the slide shaft **21** positioned at the end portion **132a** being balanced.

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Operation of Path Switching Mechanism

An operation of a guide portion **130** when the cover **10** is moved from the first opened position to the second opened position will be described. In order to move the cover **10** from the first opened position to the second opened position, it is necessary to move the slide shaft **21** from the first path **131** to the second path **132**. From the closed position illustrated in FIGS. **14A** and **14B**, the operation of moving the slide shaft **21** from the first path **131** to the second path **132** and moving the cover **10** from the first opened position to the second opened position will be described. In a state where the slide shaft **21** is positioned at the end portion **131b**, the operator moves a path switching member **125** from a position **125a** to a position **125b** illustrated in FIG. **16**, and the second path **132** is opened. Thereafter, as illustrated in FIG. **17A**, by closing the cover **10** slightly, the slide shaft **21** falls into the second path **132**, and in that state, by opening the cover **10** again, the cover **10** may be opened to the second opened position.

Operation of Path Switching Member During Opening Operation of Cover

FIG. **12** is a three-side view illustrating the interlock of the path switching mechanisms **150A** and **150B** provided on both ends in a rotation center shaft direction of the cover **10**. In image forming apparatuses **1**, the size of the cover **10** tends to increase along with the increase in corresponding sheet sizes, the enhancement of speed and the increase of functions leading to improved productivity. In the present embodiment, in order to regulate the opening angle of the cover **10** that may be large and heavy, the support arm **20** is provided on both end portions in the rotational axis direction of the cover **10**. Along therewith, guide portions **130A** and **130B** and path switching members **125A** and **125B** are provided respectively to correspond to the two support arms **20**. If slide shafts **21A** and **21B** of the two support arms **20** are positioned at different paths, deformation and damage may occur due to the twisting or distortion of the cover, or damage of the support arm **20** may be caused by the concentration of load on only one of the support arms **20**. In some cases, the cover itself may fall from the casing **1a** and the cover may be damaged. In order to prevent these drawbacks, it is desirable that the two path switching members **125A** and **125B** are interlocked when switching positions.

The interlocking of operations of the path switching members **125A** and **125B** using the interlocking portion **33** will be described with reference to FIGS. **12A** to **12C**. The present embodiment adopts a configuration in which, by mechanically connecting the path switching members **125A** and **125B** using the interlocking portion **33**, when the path switching member **125A** is moved, the path switching member **125B** is also moved infallibly in an interlocked manner. The interlocking portion **33** includes a first link **34** and a second link **35** that are mutually connected rotatably. The first link **34** includes a first connecting portion that is interlocked with the path switching member **125A**, a second connecting portion that is connected rotatably to the second link **35**, and a first supporting portion **34a** that is supported rotatably to the cover **10** between the first connecting portion and the second connecting portion. The second link **35** includes a third connecting portion that is interlocked with the path switching member **125B**, a fourth connecting portion that is connected rotatably to the second connecting portion of the first link **34**, and a second supporting portion **35a** that is supported rotatably to the cover **10** between the third connecting portion and the fourth connecting portion.

As illustrated in FIG. **12**, when the path switching member **125A** is positioned at the position **125a** (refer to FIG.

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16), the path switching member 125B is also positioned at the position 125a through the interlocking portion 33. As illustrated in FIG. 13, when the path switching member 125A is positioned at the position 125b (refer to FIG. 16), the path switching member 125B is also positioned at the position 125b through the interlocking portion 33. As described, the position of the path switching member 125B can be switched infallibly by being interlocked with the switching of position of the path switching member 125A. Operation of Path Switching Member for Cover Closing Operation, i.e., from Second Opened Position to First Opened Position or Closed Position

The closing operation of the cover 10 from the second opened position to the first opened position and to the closed position state will be described with reference to FIGS. 18 and 19. FIG. 18B illustrates a state in which the cover 10 is closed from the second opened position and the slide shaft 21 has returned from the second path 132 to the end portion 131b of the first path 131. Further, FIG. 19 illustrates a state in which the closing operation is further advanced. According to the angle of the cover 10, the path switching member 25 may be moved from the position 125b to the position 125a while the slide shaft 21 moves within the first path 131 toward the end portion 131a of the first path 131. Therefore, when an operation to close the cover 10 from the second opened position to the closed position is performed, the operator may return the path switching member 25 to the position 125a simply by closing the cover 10.

As described, in the image forming apparatus 1 of the present embodiment, according to the opening/closing angle switching operation of the cover 10, the position of the path switching member 125B may be moved in an interlocked manner with the movement of position of the path switching member 125A, such that the operation can be simplified. Therefore, not only dedicated operators but also general users can perform the switching operation, and replacement of periodically replaced units can be performed by the users.

Third Embodiment

Next, a third embodiment of the present invention will be described in detail with reference to FIGS. 20 to 28. The configuration of the present embodiment differs from the first embodiment in that a switch 225 serving as an operating portion may be locked to prevent erroneous operation. Other configurations are similar to the first embodiment, such that the same reference numbers are assigned to corresponding components, and detailed descriptions thereof are omitted. Configuration Regarding Opening and Closing of Cover

FIG. 20 illustrates a configuration regarding the opening and closing of the cover 10. The cover 10 includes a guide portion 230 having an opened shape, the switch 225, the cover rotation shaft 18, and the support arm 20. The support arm 20 includes the slide shaft 21 and the rotation shaft 22. The guide portion 230 and the slide shaft 21 are designed to be engaged, the cover rotation shaft 18 and the image forming apparatus 1 are designed to be engaged, and the rotation shaft 22 and the image forming apparatus 1 are designed to be engaged. The switch 225 is slidably attached to a switch guide 10c, which are connected through the lock portion 224 and a link 226 as illustrated in FIG. 21, wherein the link 226 is engaged rotatably with each of the switch 225 and the lock portion 224. The lock portion 224 is rotatably engaged to the cover 10 through a lock portion rotation shaft 10d. The cover 10 rotates about the cover rotation shaft 18, and the orientation of the cover 10 is determined by the slide shaft 21 and the guide portion 230.

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Transition from Closed Position to First Position

FIG. 22 illustrates a detail of a closed position of the cover 10. When the slide shaft 21 of the support arm 20 is positioned at an end portion 231a of a first path 231, the support arm 20 is approximately parallel to the cover 10, and the cover 10 will be in a closed position stored in the image forming apparatus 1. In this state, by providing a lock mechanism by which the cover 10 is locked to the image forming apparatus 1 at an arbitrary position, the slide shaft 21 will not receive any load from the cover 10. When the cover 10 in the closed position state is opened, the slide shaft 21 is guided along the first path 231 of the guide portion 230 and moves from the end portion 231a to an end portion 231b, by which the cover 10 is transited to the first opened position illustrated in FIG. 23. When the slide shaft 21 of the support arm 20 is positioned at the end portion 231b of the first path 231, the support arm 20 is in a balanced state by the rotation shaft 22 and the slide shaft 21 positioned at the end portion 231b. The cover 10 is in a first opened position state by the cover rotation shaft 18 and the slide shaft 21 positioned at the end portion 231b being balanced.

Transition from First Opened Position to Second Opened Position

As illustrated in FIG. 23, in a state where the cover 10 is in a first opened position, the slide shaft 21 is regulated from being guided toward a second path 232 by the lock portion 224 at a branch portion 230a of the guide portion 230. When the switch 225 is operated toward an arrow D1 direction from this state, as illustrated in FIG. 24, a retained state is realized by a switch pawl portion 225a being engaged to a projected portion 10b of the cover 10. Further, by operating the switch 225, a moment of rotation in a D3 direction occurs about the lock portion rotation shaft 10d to the lock portion 224 through the link 226, and the lock portion 224 moves to a position where the slide shaft 21 is not regulated. A first spring 227 is a compression coil spring which is disposed to constantly urge the switch 225 toward the direction D2 to ensure the regulated state of the lock portion 224, and it is desirably provided. Alternatively, as a unit other than the spring, a configuration may be adopted to elastically engage the switch 225 and the cover 10.

A second spring 228 is provided with the aim to aid the operation of the switch 225, by which a moment of rotation in a direction D3 occurs to the lock portion 224, which applies a force in a D1 direction to the switch 225 through the link 226, and it is preferably provided to reduce the force necessary for operation. In this state, the cover 10 and the support arm 20 are in a balanced state and stopped at a first opened position, as illustrated in FIG. 28. By lifting the cover 10 slightly by which the slide shaft 21 is separated from the end portion 231b, the state of balance is released and the slide shaft 21 is guided toward the second path 232.

Now, the cover 10 will not be guided to the second path 232 immediately after switching the lock portion 224, and the balanced state is maintained. That is, the lock portion 224 merely blocks the second path 232, and the load of the cover 10 is received by the end portion 231b provided on the first path 231. Thereby, the only operating force when operating the lock portion 224 is the load that occurs when engaging the switch pawl portion 225a to the projected portion 10b of the cover 10. By adopting such configuration, the operating force that the user exerts during operation may be reduced.

As illustrated in FIG. 25, the slide shaft 21 guided to the second path 232 reaches an end portion 232a, and the support arm 20 and the cover 10 are stopped in the balanced state. Thus, transition to the second opened position is

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completed. When the slide shaft 21 of the support arm 20 is positioned at the end portion 232a of a second guide portion, the support arm 20 is in a balanced state by the rotation shaft 22 and the slide shaft 21 positioned at the end portion 232a. The cover 10 is in a second opened state by the cover rotation shaft 18 and the slide shaft 21 positioned at the end portion 232a being balanced.

Transition from First and Second Opened Position to Closed Position

When the user closes the cover 10 from the first opened position illustrated in FIG. 23, the slide shaft 21 is guided by the guide portion 230 to move to the end portion 231a and to the closed position. Further, if the user similarly closes the cover 10 from the second opened position illustrated in FIG. 25, the slide shaft 21 is guided by the guide portion 230 to be passed through the branch portion 230a and abut against the lock portion 224. When the cover 10 is closed further from this state, as illustrated in FIG. 26, a moment of rotation in a D4 direction occurs to the lock portion 224, along with which the switch 225 and the link 226 are returned to their initial positions in an interlocked manner, and the slide shaft 21 is guided by the guide portion 230 to the end portion 231a and transited to the closed position. As described, since the cover 10 may be transited to the closed position by simply closing the cover 10, the action of the user operating the switch 225 before closing the door can be omitted.

Configuration of Measures Against Erroneous Operation in Second Opened Position

In the present embodiment, in a state where the cover 10 is in the second opened position illustrated in FIG. 25, if the switch 225 is operated in the D2 direction toward the returning direction, a moment of rotation in the D3 direction as illustrated in FIG. 26 occurs, such that the lock portion 224 will not return to the initial position. If it is assumed that the switch 225 has returned to the initial position, the state of FIG. 27 is realized. That is, the lock portion 224 will be positioned in midway of the guide portion 230 forming a wall. In that case, if the user closes the cover 10 without performing an operation to return the switch, the slide shaft 21 and the lock portion 224 will collide against each other and may be damaged, so that the above-mentioned configuration is adopted to prevent such a state.

As described, according to the image forming apparatus 1 of the present embodiment, when switching the cover 10 from the first opened position to the second opened position, the switching is enabled through a switching operation. If the user operates the switch at an unintended timing, the switch 225 will not be returned, and the cover 10 is moved to the closed position only through the operation of closing the cover 10, such that that intuitive and safe operation is enabled.

Fourth Embodiment

Next, a fourth embodiment of the present invention will be described in detail with reference to FIGS. 29A to 31. The configuration of the present embodiment differs from the first embodiment in that a function of retreating a fixing member 40 from a first state is provided. Other configurations are similar to the first embodiment, such that the same reference numbers are assigned to corresponding components, and detailed descriptions thereof are omitted.

Configuration Related to Opening and Closing of Cover

The support arm 20 includes the slide shaft 21 connecting the support arm 20 and the cover 10, and the rotation shaft 22 connecting the support arm 20 and the image forming

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apparatus 1. The cover 10 is provided with a guide portion 330 that guides the support arm 20, and the angle, i.e., position and orientation, of the support arm 20 is determined by the position of the slide shaft 21 on the guide portion 330, and simultaneously, the opening angle of the cover 10 is determined.

FIG. 29A is an enlarged view illustrating from a front side of the apparatus the cover 10 in a state opened to the first angle. As illustrated in FIG. 29A, the fixing member 40 is provided to retain the cover 10 at the first angle. The fixing member 40 is provided on the cover 10 and the fixing member 40 contacts the slide shaft 21 guided by the guide portion 330 by a support surface 43, and retains the slide shaft 21 at the position illustrated in FIG. 29A. The fixing member 40 is supported on the cover 10 in a manner rotatable from the state illustrated in FIG. 29A, i.e., first state, about a fixing member rotation shaft 41, and the fixing member 40 is retained by a rotation stop portion not shown so as not to rotate in a counterclockwise direction. Accordingly, the support arm 20 is in a state balanced by the rotation shaft 22 and the slide shaft 21 in contact with the support surface 43, and the cover 10 is also retained at the first angle by the cover rotation shaft 18 and the slide shaft 21 in contact with the support surface 43 being balanced.

FIG. 29B is an enlarged view illustrating from a front side of the apparatus a state in which the fixing member 40 is moved and the cover 10 is transited from the first angle to the second angle. If the slide shaft 21 of the support arm 20 is moved to an end portion of the guide portion 330 illustrated in FIG. 29B, the support arm 20 is in a balanced state by the rotation shaft 22 and the slide shaft 21. The cover 10 is retained at the second angle by the cover rotation shaft 18 and the slide shaft 21 positioned at the end portion of the guide portion 330 being balanced. The fixing member 40 has the state of FIG. 29B set as the second state.

In order to open the cover 10 from the first angle to the second angle, that is, in order to move the slide shaft 21, the rotation stop portion, i.e., rotation lock, not shown of the fixing member 40 should be moved by the operator. In this state, the rotation stop portion may be movably supported on the cover 10 or may be removably disposed on the cover 10. As described above, by moving the rotation stop portion, the fixing member 40 will be in a state rotatable in the counterclockwise direction, and along with the rotation of the fixing member 40, the slide shaft 21 is moved within the guide portion 330 to a position where the cover 10 is retained at the second angle. In the present embodiment, after the slide shaft 21 moves beyond the fixing member 40 toward the end portion of the guide portion 330 illustrated in FIG. 29B, the fixing member 40 is designed be returned automatically to the first state illustrated in FIG. 29A.

Closing Operation from Second Position

FIG. 30A is an enlarged view illustrating from a front side of the apparatus a movement of the slide shaft 21 when the cover 10 is closed from the second angle to the first angle, and FIG. 30B is an enlarged view illustrating from the front side of the apparatus a movement of the slide shaft 21 and the fixing member 40 when the cover 10 is closed from the second angle to the first angle. Actually, when the slide shaft 21 is moved, the movement of the slide shaft 21 is interlocked with the movement in which the cover 10 is closed from the second angle to the first angle, such that the angle is further closed toward the body of the image forming apparatus 1 than the opening angle of the cover 10 illustrated in FIG. 30A. However, the movement of the slide shaft 21 and the fixing member 40 is relevant to the present embodiment, such that further movement of the cover 10 is omitted.

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In the present embodiment, after the slide shaft 21 passes the fixing member 40 toward the end portion of the guide portion 330 illustrated in FIG. 29B, the fixing member 40 is designed to be automatically returned to the state illustrated in FIG. 29A. In this case, in a general conventional configuration, as illustrated in FIG. 30A, the slide shaft 21 comes into contact with a non-support surface 44 of the fixing member 40 and the fixing member 40 will not move, such that the slide shaft 21 is blocked in this position. If the cover 10 is forcibly closed from this state, the fixing member 40, the support arm 20, or the cover 10 may be deformed or damaged.

Therefore, according to the present embodiment, as illustrated in FIG. 30B, the fixing member 40 is supported rotatably further in the clockwise direction from the state of FIG. 30A. Thereby, when the slide shaft 21 moves from the state of FIG. 30A toward the closing direction of the cover 10, the non-support surface 44 is pushed by the slide shaft 21 abutted thereagainst, and the fixing member 40 rotates in the clockwise direction in an interlocked manner. That is, since the movement of the slide shaft 21 is not regulated by the fixing member 40, drawbacks such as deformation and damage may be suppressed even if the cover 10 is closed from the state of FIG. 30A. Moreover, as illustrated in FIG. 30B, the fixing member 40 rotates in the clockwise direction and the slide shaft 21 moves to a position where the slide shaft 21 and the non-support surface 44 are separated. Thereafter, the fixing member 40 is caused to automatically rotate in the counterclockwise direction to the first state illustrated in FIG. 30A by an elastic member not shown, such as a spring. Thereby, the user is not required to return the fixing member 40 to the first state illustrated in FIG. 30A, and the cover 10 can be retained at the first angle without fail when the cover 10 is opened for the next time from the closed state.

As described above, according to the image forming apparatus 1 of the present embodiment, the fixing member 40 in the second state automatically moves to the first state, such that the user is required to simply close the cover 10, according to which the operability may be enhanced.

According to the image forming apparatus 1 of the above-mentioned embodiment, a case in which the guide portion 330 is linear has been described, but the present technique is not limited thereto. For example, as illustrated in FIG. 31, similar to the first to third embodiments, the first path 331 and the second path 332 may adopt stepped shapes. In this case, when the slide shaft 21 is positioned at an end portion 331a of the first path 331, the support arm 20 is in a balanced state by the rotation shaft 22 and the slide shaft 21 positioned at the end portion 331a. The cover 10 is retained at the first angle by the cover rotation shaft 18 and the slide shaft 21 positioned at the end portion 331a being balanced. According to the configuration illustrated in FIG. 31, when the cover 10 is opened to the first angle from the state closed with respect to the image forming apparatus 1, the fixing member 40 closes a connecting portion between the first path 331 and the second path 332 such that the slide shaft 21 is prevented from entering the second path 332. That is, the slide shaft 21 is guided toward the end portion 331a by a guide surface 43b of the fixing member 40. Thereby, in a state where the cover 10 is retained at the first angle, the slide shaft 21 maintains a state where load is not applied to the fixing member 40. By adopting this configuration, it becomes possible to reduce the stiffness and strength of the fixing member 40, that is, it becomes possible

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to use a small member formed of a lightweight material, such that the image forming apparatus 1 can be downsized with reduced weight.

According to the image forming apparatus 1 of the above-mentioned embodiment, after reaching the second state and the slide shaft 21 has passed the fixing member 40 toward the end portion of the guide portion 330 illustrated in FIG. 29B, the fixing member 40 is caused to return automatically to the first state illustrated in FIG. 29A. However, the present technique is not limited thereto, and for example, similar to the first to third embodiments, the fixing member 40 may be caused to maintain the second state after the slide shaft 21 has passed the fixing member 40 toward the end portion of the guide portion 330 illustrated in FIG. 29B. In that case, when closing the cover 10 from the second angle to the first angle, the slide shaft 21 moves within the guide portion 330 and abuts against a returning surface 42 of the fixing member 40. Further, by the fixing member 40 rotating in the clockwise direction in an interlocked manner with the movement of the slide shaft 21, the fixing member 40 is transited to the first state illustrated in FIG. 29A. Then, when the user returns the rotation stop portion not shown, the fixing member 40 will no longer rotate in the counterclockwise direction from the first state illustrated in FIG. 29A, and the cover 10 may be retained at the first angle. Of course, by closing the cover 10 further from this state toward the image forming apparatus 1, the cover 10 may be closed completely. However, even according to this configuration, the rotation stop portion not shown may be returned to the clockwise direction due to external forces applied by an operator or by vibration and collision, before the operation of closing the cover 10 from the second angle to the first angle as illustrated in FIG. 30A is performed. In that case, according to the present embodiment, when the slide shaft 21 moves from the state of FIG. 30A toward the closing direction of the cover 10, the non-support surface 44 is pressed by the slide shaft 21 in contact therewith, and the fixing member 40 rotates in the clockwise direction in an interlocked manner, such that the cover 10 may be closed.

Fifth Embodiment

Next, a fifth embodiment of the present invention will be described in detail with reference to FIG. 32. The configuration of the present embodiment differs from the fourth embodiment in that it has a function to retreat from the first state by having a tip of the fixing member rotate. Other configurations are similar to the fourth embodiment, such that the same reference numbers are assigned to corresponding components, and detailed descriptions thereof are omitted.

FIG. 32A is an enlarged view illustrating from a front side of the apparatus a movement of the slide shaft 21 in a state where the cover 10 according to the present embodiment is closed from a second angle to a first angle. FIG. 32B is an enlarged view illustrating from a front side of the apparatus a movement of the slide shaft 21 and the fixing member 40 in a state where the cover 10 is closed from the second angle to the first angle. As illustrated in FIG. 32A, the movable member 60 is provided on the fixing member 40. The movable member 60 is supported rotatably about a rotation shaft 61 with respect to the fixing member 40, and rotation thereof is restricted by a rotation stopper not shown so as not to rotate in the counterclockwise direction with respect to the fixing member 40 from the state of FIG. 32A. Thereby, when the slide shaft 21 moves from the state of FIG. 32A toward the direction closing the cover 10, the movable

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member 60 is pressed by the slide shaft 21 in contact therewith, and the movable member 60 is rotated in an interlocked manner in the clockwise direction. That is, since the movement of the slide shaft 21 is not restricted by the movable member 60, the cover 10 may be closed from the state of FIG. 32A.

Furthermore, as illustrated in FIG. 32B, the movable member 60 rotates in the clockwise direction, and the slide shaft 21 is moved to a position where the slide shaft 21 and the movable member 60 are separated. Thereafter, the movable member 60 is rotated in the counterclockwise direction with respect to the fixing member 40 automatically to the position illustrated in FIG. 32A by an elastic member not shown, such as a spring. Thereby, the user is not required to return the movable member 60 to the state of FIG. 32A, and the cover 10 can be retained at the first angle without fail when the cover 10 is opened for the next time from the closed state.

According to the present invention, it becomes possible to simply the operability when opening the cover in two stages.

OTHER EMBODIMENTS

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of Japanese Patent Application No. 2022-177474, filed Nov. 4, 2022 which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. An image forming apparatus comprising:

an image forming unit configured to form an image on a sheet;

a casing accommodating the image forming unit;

a door configured to be opened and closed with respect to the casing, and configured to be moved to a first opened position and a second opened position, the first opened position being a position where the door is inclined by a first angle with respect to a closed position, where the casing is closed and a sheet conveyance path is formed, to expose the sheet conveyance path, the second opened position being a position where the door is inclined by a second angle that is greater than the first angle with respect to the closed position to expose the sheet conveyance path;

a connecting member configured to be interlocked with an opening and closing operation of the door, and connected to the casing and the door;

a guide member provided on the door, the guide member including a first guide configured to guide the connecting member in a case that the door is moved between the closed position and the first opened position, and a second guide branched from the first guide and configured to guide the connecting member in a case that the door is moved to the second opened position, wherein the connecting member is configured to retain the door at the first opened position by engaging an end of the first guide and to retain the door at the second opened position by engaging an end of the second guide; and

an operating portion configured to move, through an operation by a user and in a state where the door is retained at the first opened position by the connecting member engaging the end of the first guide, from a

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regulating position where the connecting member is regulated from moving from the first guide to the second guide, to an allowing position where the connecting member is allowed to move from the first guide to the second guide,

wherein, in a state where the operating portion is at the allowing position, the door is configured to be retained at the second opened position by movement of the connecting member from the first guide to the second guide.

2. The image forming apparatus according to claim 1, wherein the connecting member is configured to be switched between a first state in which the door is rotatable from the closed position to the first opened position, and a second state in which the door is rotatable at least from the first opened position to the second opened position, and

wherein the operating portion is configured to switch the connecting member between the first state and the second state.

3. The image forming apparatus according to claim 2, further comprising:

a regulation portion configured to be interlocked with the operating portion,

wherein the connecting member includes a guided portion configured to be guided by the guide member,

wherein the regulation portion is configured to regulate a movement of the guided portion,

wherein, in a state where the guided portion is engaged with the first guide and the operating portion is at the regulating position, the regulation portion is configured to regulate the guided portion from entering to the second guide, by which the connecting member is in the first state, and

wherein, in the state where the operating portion is at the allowing position, the connecting member is in the second state by the first guide and the second guide being communicated.

4. The image forming apparatus according to claim 3, wherein the connecting member is configured to be switched from the first state to the second state by an operation of the operating portion being moved from the regulating position to the allowing position, and wherein the connecting member is configured to be switched from the second state to the first state by rotating the door from the second opened position to the closed position.

5. The image forming apparatus according to claim 2, further comprising:

a regulation portion configured to be interlocked with the operating portion,

wherein the connecting member includes a guided portion configured to be guided by the guide member,

wherein the regulation portion includes a contact portion configured to intersect a trajectory of the guided portion in the first guide in a state where the connecting member is in the second state, and

wherein, while the door moves from the second opened position to the closed position, the guided portion comes in contact with the contact portion to move the regulation portion from the allowing position to the regulating position.

6. The image forming apparatus according to claim 2, further comprising:

a regulation portion configured to be interlocked with the operating portion, and

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a retaining portion configured to retain the regulation portion such that the connecting member is maintained in the first state,

wherein the regulation portion is released from being retained by the retaining portion by an operation of the operating portion being moved from the regulating position to the allowing position.

7. The image forming apparatus according to claim 3, wherein the guide member includes a receiving portion configured to receive a load of the door acting on the guided portion in a state where the door is positioned at the first opened position.

8. The image forming apparatus according to claim 1, wherein the operating portion is configured to move to the regulating position and the allowing position.

9. The image forming apparatus according to claim 1, wherein the operating portion is disposed at a portion of the door other than a portion forming an external wall of the casing in the closed position.

10. The image forming apparatus according to claim 1, wherein the door forms a side wall portion of the casing in the closed position, and is configured to rotate about a rotation shaft in a horizontal direction at a lower part of the door.

11. An image forming apparatus comprising:

an image forming unit configured to form an image on a sheet;

a casing accommodating the image forming unit;

a door configured to be opened and closed with respect to the casing, and configured to be moved to a first opened position and a second opened position, the first opened position being a position where the door is inclined by a first angle with respect to a closed position, where the casing is closed and a sheet conveyance path is formed, to expose the sheet conveyance path, the second opened position being a position where the door is inclined by a second angle that is greater than the first angle with respect to the closed position to expose the sheet conveyance path;

a connecting member connected to the casing and the door;

a guide member provided on the door, the guide member including a first guide configured to guide a guided portion of the connecting member in a case that the door is moved between the closed position and the first opened position, and a second guide branched from the

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first guide and configured to guide the guided portion in a case that the door is moved to the second opened position, wherein the connecting member is configured to retain the door at the first opened position by engagement of the guided portion of the connecting member and an end of the first guide and to retain the door at the second opened position by engagement of the guided portion of the connecting member and an end of the second guide; and

a regulating portion configured to move, in a state where the door is retained at the first opened position by the guided portion of the connecting member engaging the end of the first guide, from a regulating position where the guided portion is regulated from moving from the first guide to the second guide by contact with the regulating portion, to an allowing position where the guided portion is allowed to move from the first guide to the second guide,

wherein, in a case where the door is opened in a state where the regulating portion is at the regulating position, the guided portion is regulated from moving from the first guide to the second guide by the regulating portion, and

wherein, in a case where the door is opened in a state where the regulating portion is at the allowing position, the guided portion moves from the first guide to the second guide to allow the door to be at the second opened position.

12. The image forming apparatus according to claim 11, further comprising:

an operating portion configured to be operated by a user to move the regulating portion.

13. The image forming apparatus according to claim 11, wherein the regulating portion is configured to move independently from the connecting member.

14. The image forming apparatus according to claim 11, wherein the regulating portion includes a contact portion configured to intersect a trajectory of the guided portion in the first guide in a state where the door is at the second opened position, and

wherein, while the door moves from the second opened position to the closed position, the guided portion comes in contact with the contact portion to move the regulating portion from the allowing position to the regulating position.

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